

2.1) LINE FREQUENCY CONTROL- (variable frequency constant voltage)

$$\cancel{N_s} \rightarrow \cancel{N} = \frac{120 \cancel{f}}{P} \rightarrow \cancel{N}$$

(synchronous speed) (rotor speed)

Why Torque decreasing?

$$E = 4.44 f \phi N \text{ Volts}$$

$$E \propto f \phi$$

$$\phi \propto \frac{E}{f}$$

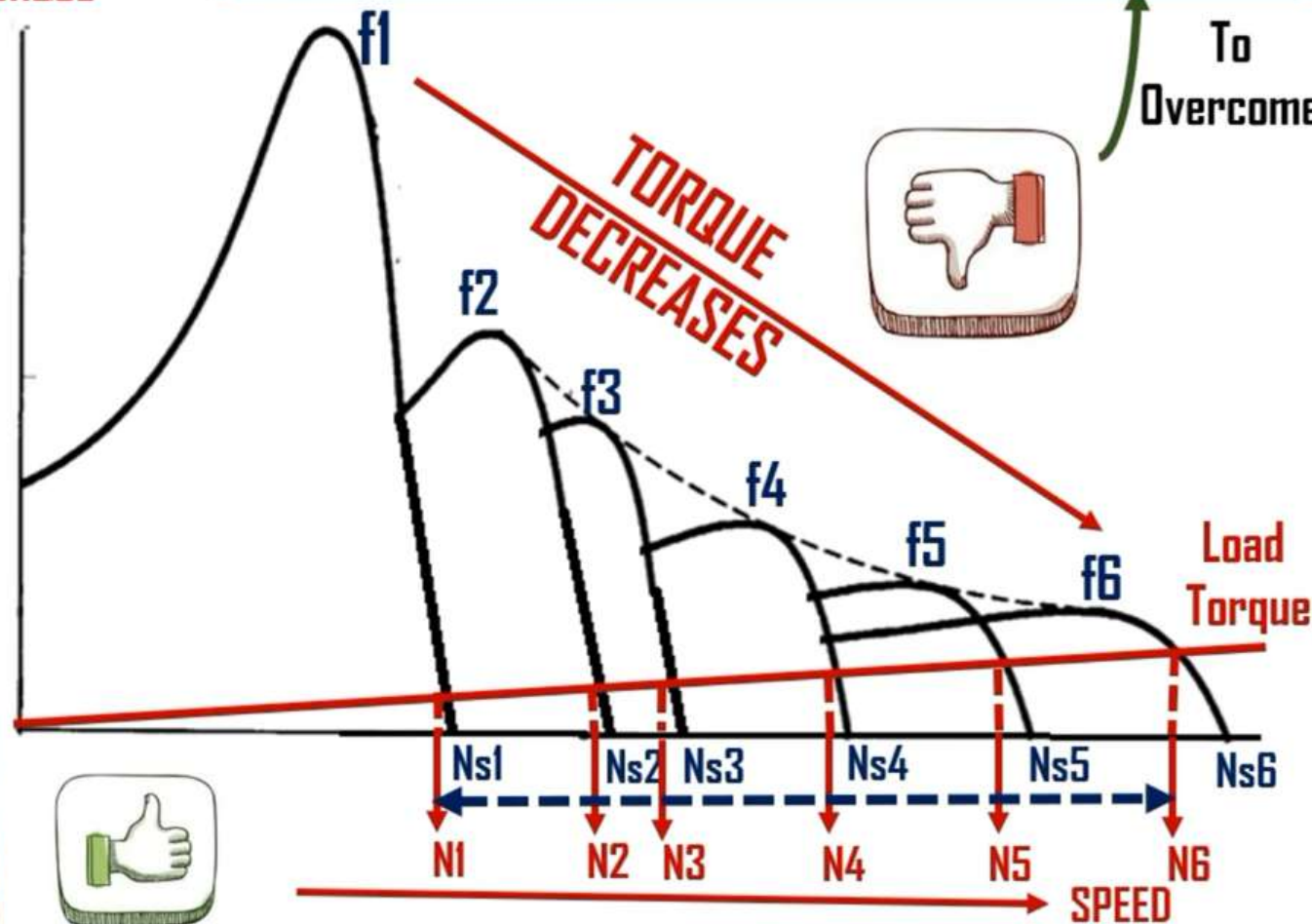
High Saturation!!

$$\downarrow T \propto \uparrow \downarrow \phi \propto \frac{V}{f}$$

(Voltage drop in R_1 & X_1)

VERY IMPORTANT POINT
If, $\phi(\text{CONST}) \propto T(\text{CONST}) \propto \frac{V}{f}(\text{CONSTANT})$

TORQUE



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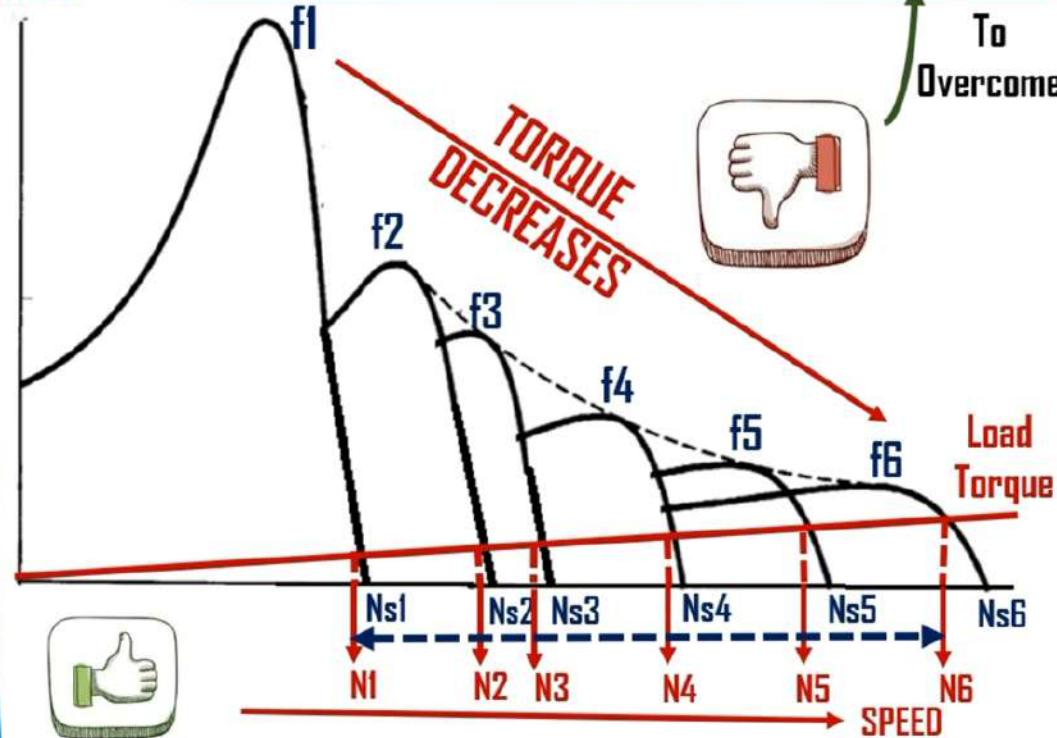
High Saturation!!

$$\downarrow T \propto \downarrow \phi \propto \frac{V}{f}$$

(Voltage drop in R_l & X_l are small then, $E \approx V$)

VERY IMPORTANT POINT
If, $\phi(\text{CONST}) \propto T(\text{CONST}) \propto \frac{V}{f}(\text{CONSTANT})$

TORQUE



$\text{SLIP} \propto (N_s - N) = \text{LOW}; \% \eta = \text{HIGH} \text{ \& \; VERY WIDER SPEED VARIATION}$

2.2) V/F CONTROL

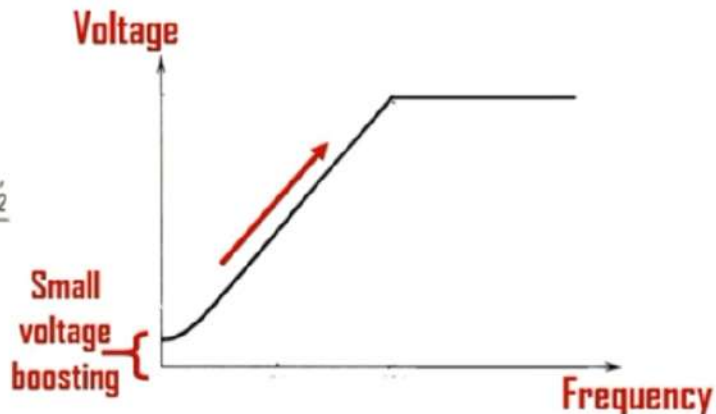
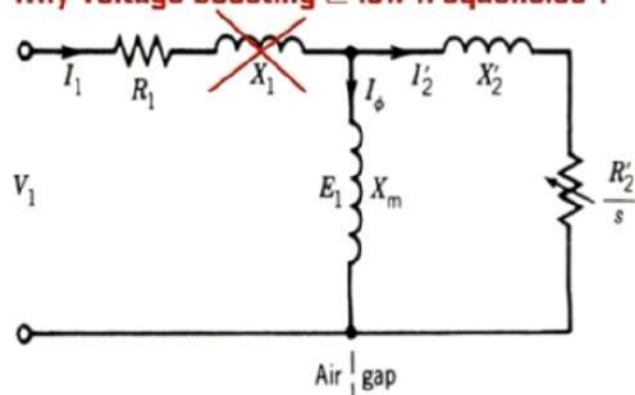
$$\phi(\text{CONST}) \propto T(\text{CONST}) \propto \frac{V}{f}(\text{CONSTANT}) \quad \text{EQUATION (I)}$$

1) HOW TO MAINTAIN V/f AS CONSTANT? $\frac{V}{f} \rightarrow \text{CONSTANT}$

2) HOW FREQUENCY CHANGES? "Based on the desired rotor speed N^* , we need to change the frequency by which the synchronous speed N_s and hence the actual rotor speed N

3) HOW TO CHANGE THE VOLTAGE WITH RESPECT TO FREQUENCY?

Why voltage boosting @ low frequencies?



2.2) V/F CONTROL

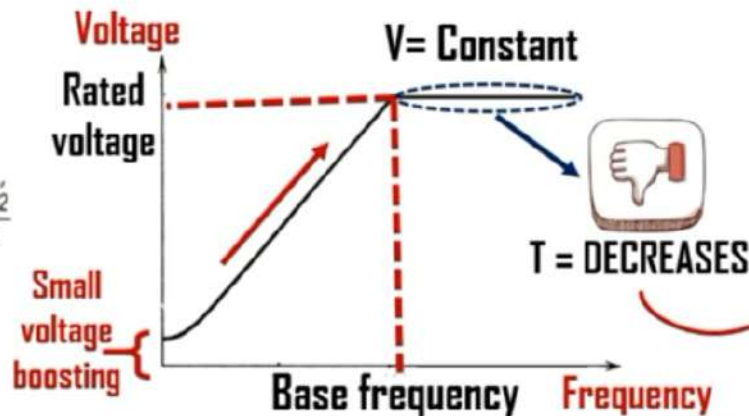
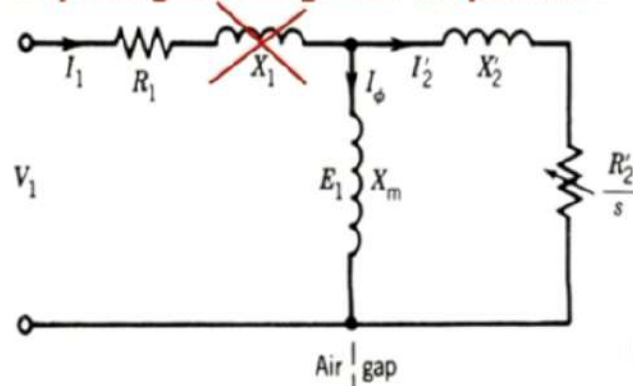
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1) HOW TO MAINTAIN V/f AS CONSTANT? $\frac{V}{f} \rightarrow \text{CONSTANT}$

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Why voltage boosting @ low frequencies?



We need to sacrifice this to avoid Insulation Breakdown in stator windings!

TORQUE - SPEED CHARACTERISTICS - V/F CONTROL

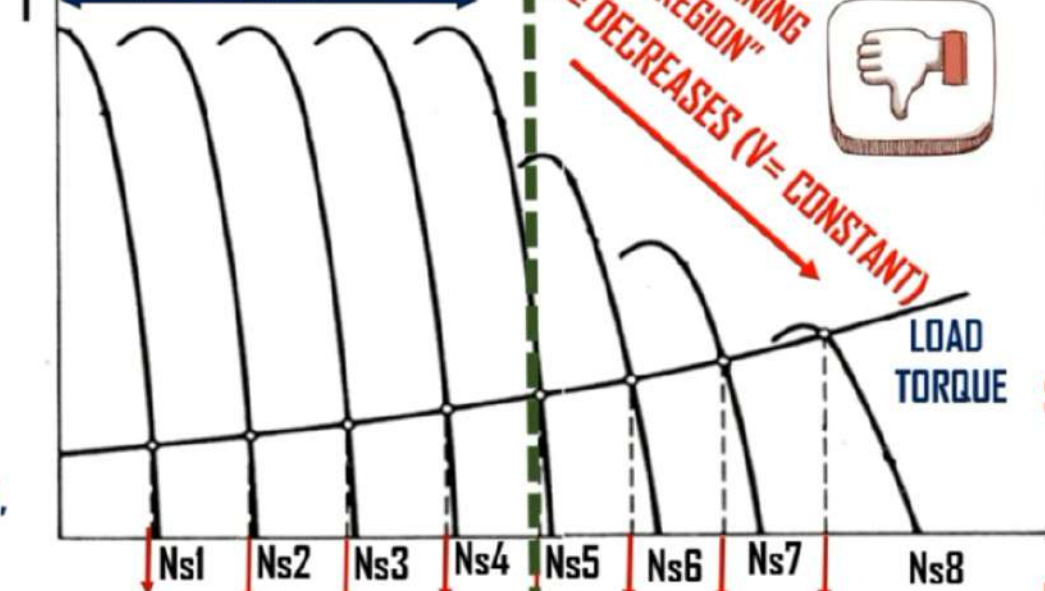


VERY WIDE SPEED VARIATION & THE
 $SLIP \propto (N_s - N) = \text{LOW}$; SO,
 EFFICIENCY = HIGH

CONSTANT TORQUE REGION



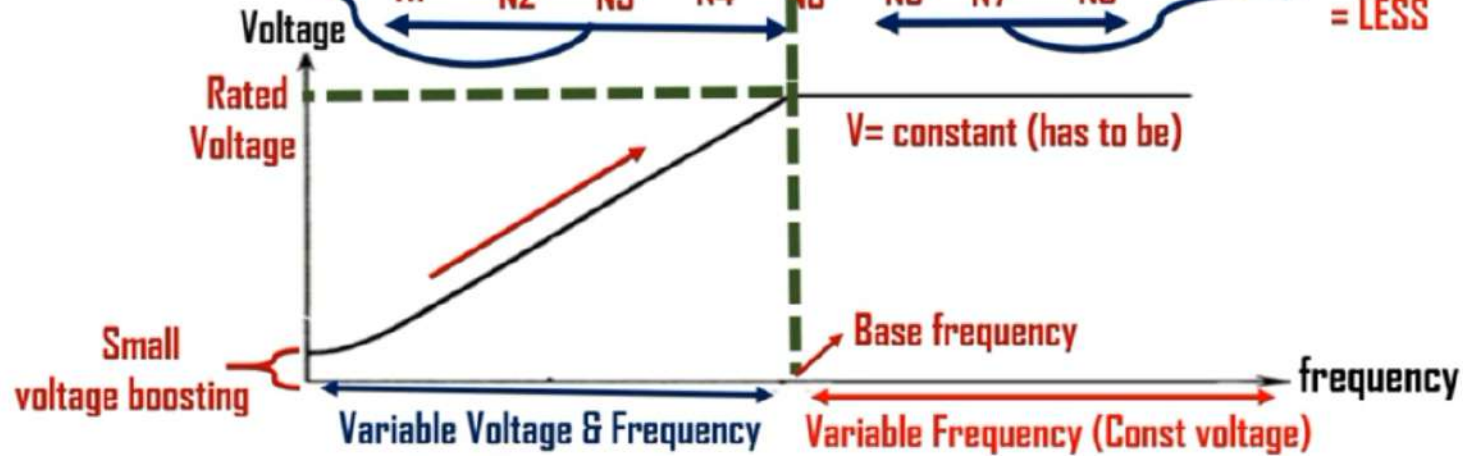
$T = \text{CONSTANT}$



"FLUX WEAKENING REGION"
 $T = \text{DECREASES (V = CONSTANT)}$



NO STIFFNESS;
 $SLIP \propto (N_s - N) = \text{HIGH}$; SO,
 EFFICIENCY = LESS



Small voltage boosting

Rated Voltage

$V = \text{constant (has to be)}$

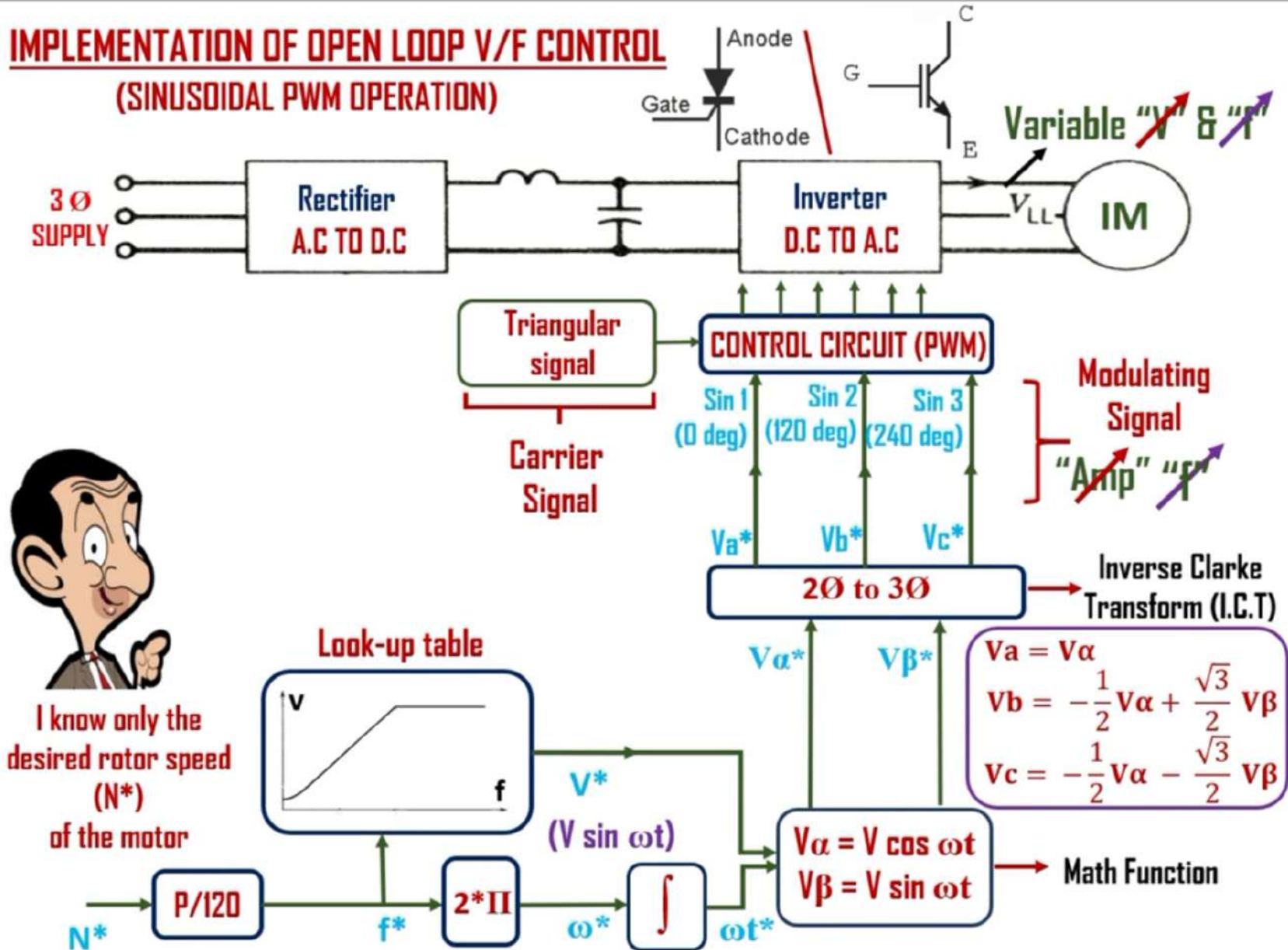
Base frequency

frequency

Variable Voltage & Frequency

Variable Frequency (Const voltage)

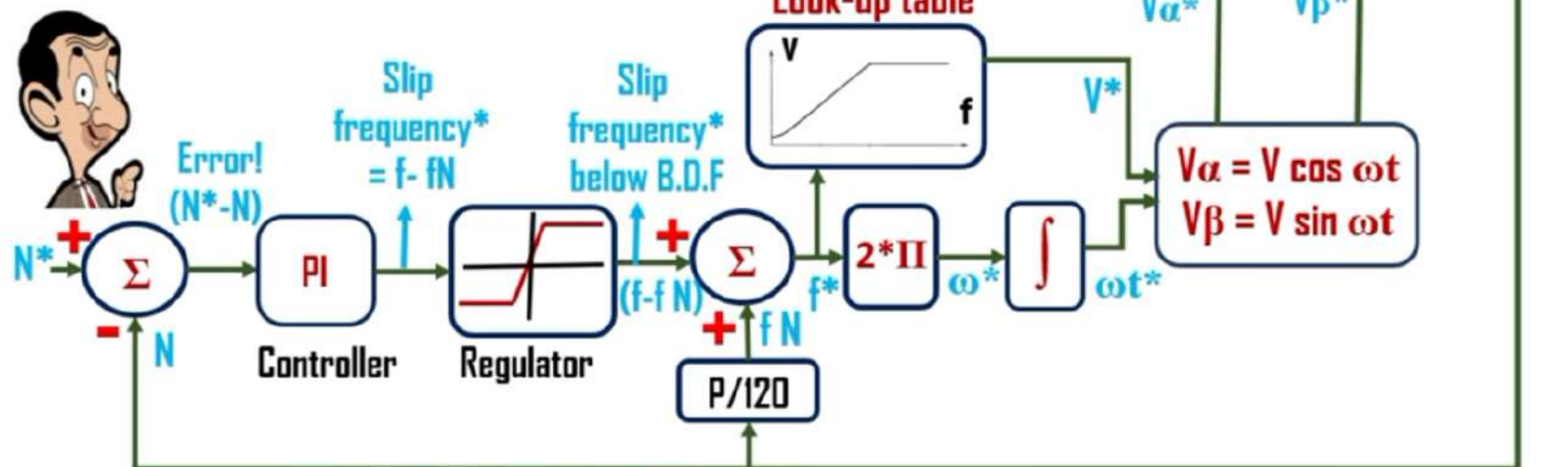
IMPLEMENTATION OF OPEN LOOP V/F CONTROL (SINUSOIDAL PWM OPERATION)



IMPLEMENTATION OF CLOSED LOOP V/F CONTROL (PWM)

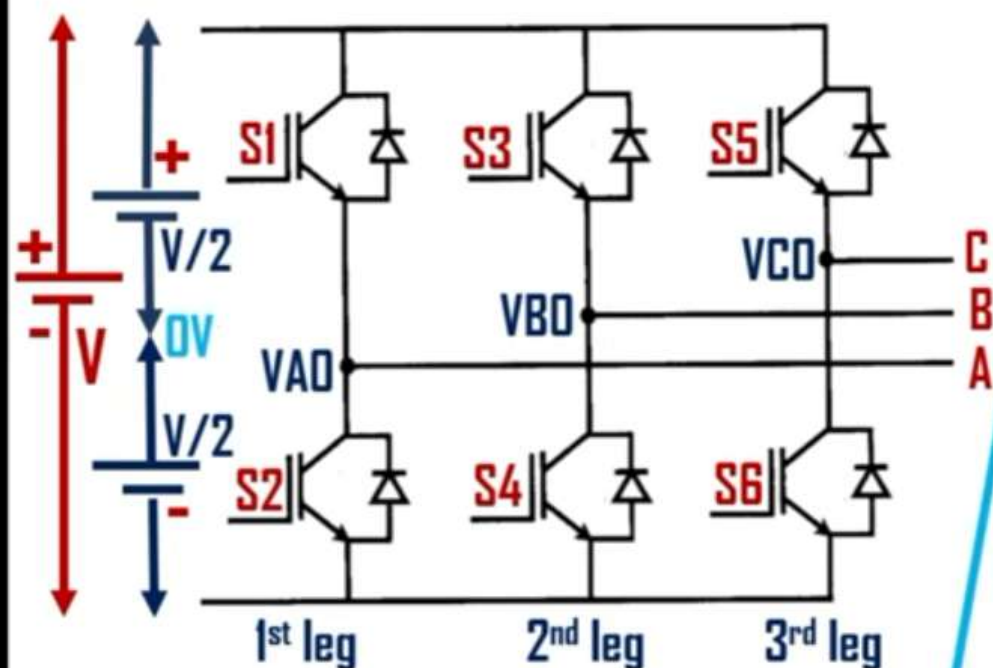
B. D. F $\rightarrow$ Break Down Frequency
(freq @ maximum torque developed)

I. C. T $\rightarrow$ Inverse Clarke Transform



VOLTAGE SOURCE INVERTER FED 3 PHASE IM (SINUSOIDAL PWM)

HOW VOLTAGE (MAGNITUDE) & FREQUENCY CONTROL IS DONE USING PWM OPERATION ?



If we control the pole voltages (by controlling the gate voltages of IGBTs switches), then it is possible to control the magnitude & frequency of line voltages

VAO, VBO, VCO = POLE VOLTAGES

(You can directly control these pole voltage by controlling the gate voltages of IGBT switches!)

For example:

- When **S1 (upper)** turned ON then, $VAO \rightarrow +V/2$
- When **S2 (lower)** turned ON then, $VAO \rightarrow -V/2$
- When **S3 (upper)** turned ON then, $VBO \rightarrow +V/2$
- When **S4 (lower)** turned ON then, $VBO \rightarrow -V/2$
- When **S5 (upper)** turned ON then, $VCO \rightarrow +V/2$
- When **S6 (lower)** turned ON then, $VCO \rightarrow -V/2$

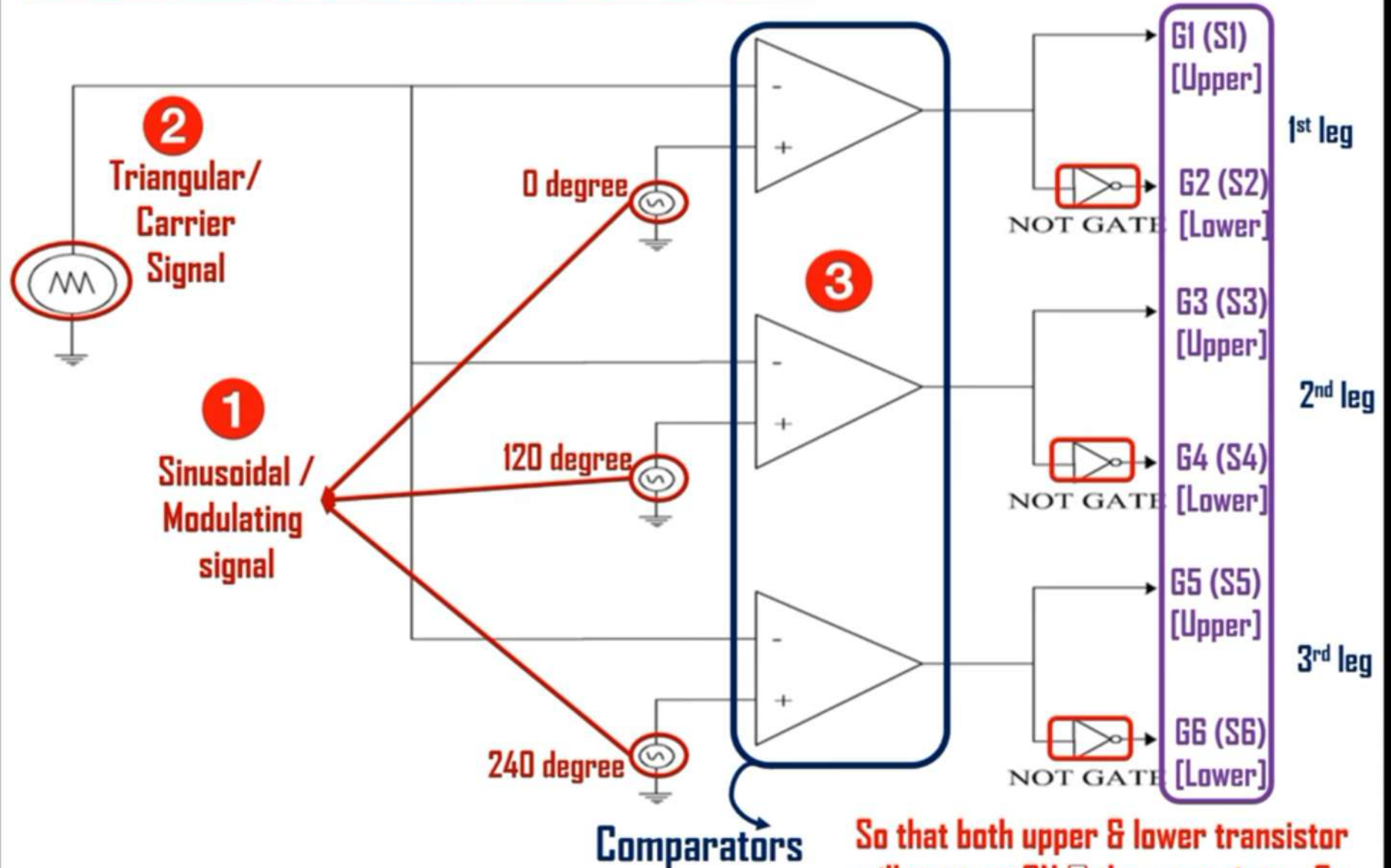
But in V/F control, what we need to control is "magnitude" & "frequency" of line voltages (VAB, VBC, VCA)

$$VAB \text{ (0 DEG Phase shift)} = VAO - VBO$$

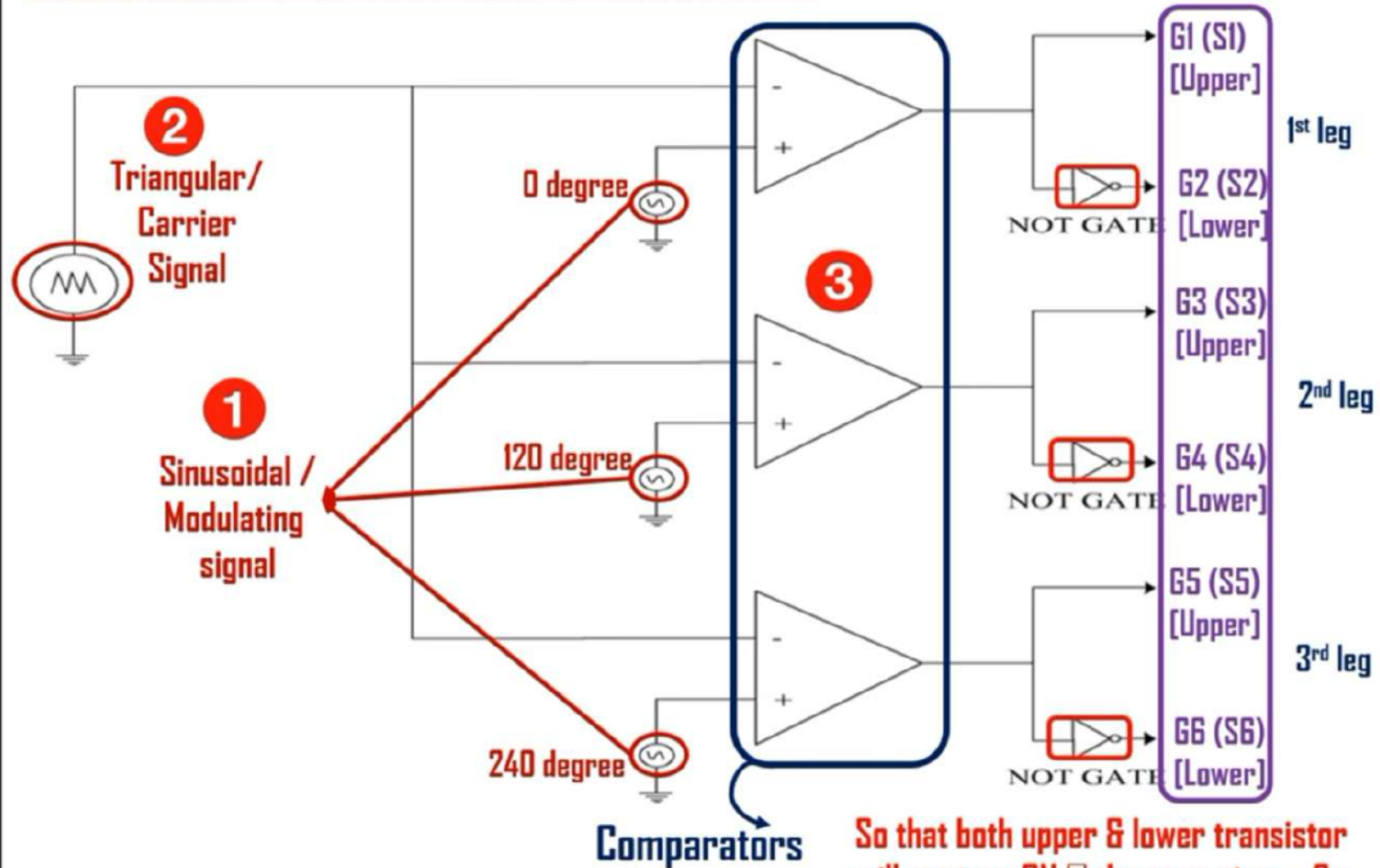
$$VBC \text{ (120 DEG Phase shift)} = VBO - VCO$$

$$VCA \text{ (240 DEG Phase shift)} = VCO - VAO$$

HOW CONTROLLED GATE VOLTAGES ARE GENERATED IN PWM?



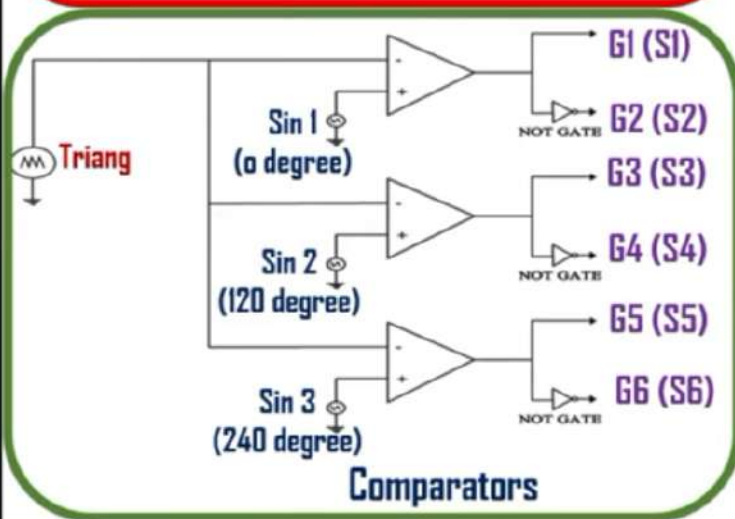
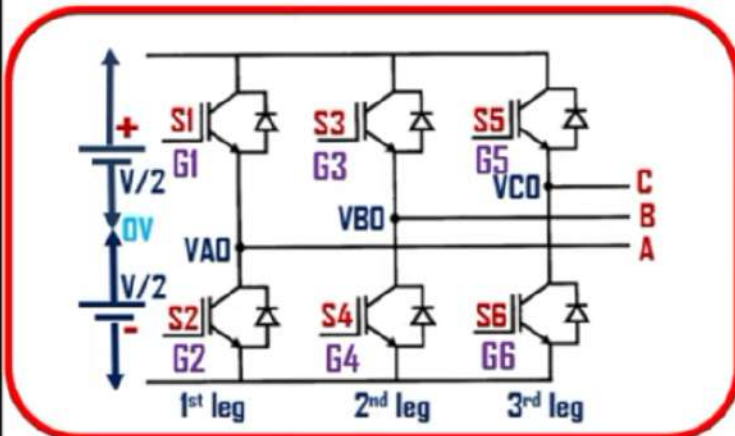
HOW CONTROLLED GATE VOLTAGES ARE GENERATED IN PWM?



Comparators

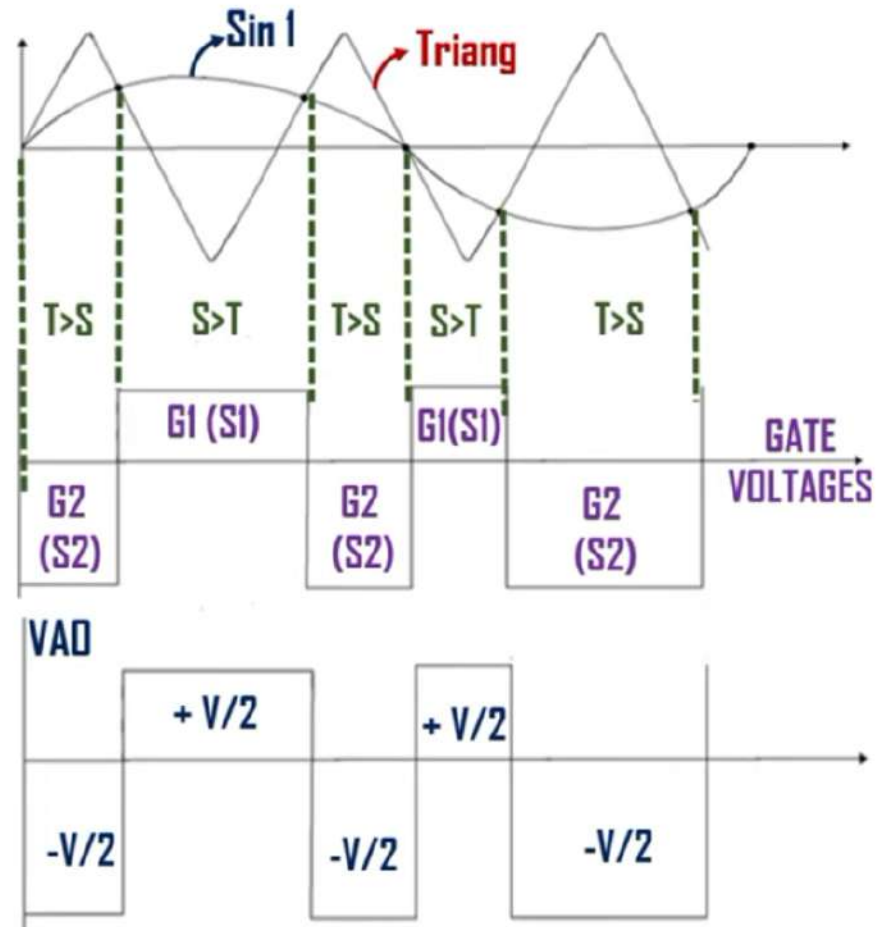
So that both upper & lower transistor will not turn ON @ the same time – So Short circuit will not happen!

POWER CIRCUIT



PWM CONTROL CIRCUIT

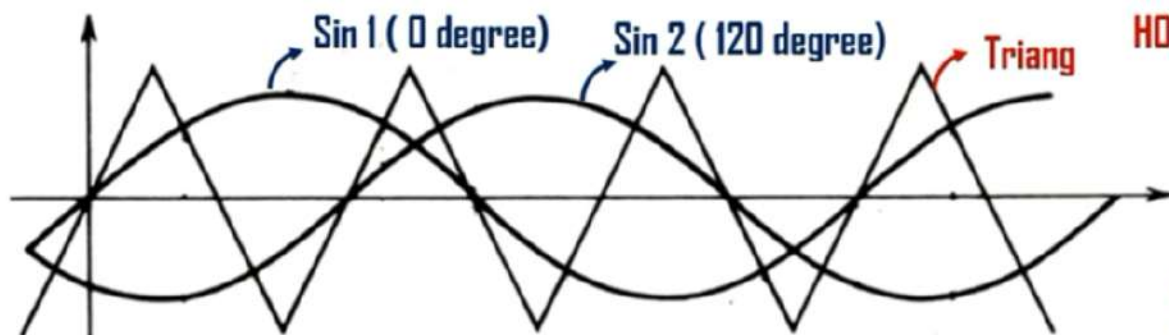
Lets analyze how gate voltages are generated for single sinusoidal signal and triangular signal



-- What we need to control is magnitude & frequency of line voltages !

So lets analyze for single line voltage VAB

$$V_{AB} = V_{AO} - V_{BO}$$



HOW TO CONTROL THE FREQUENCY OF LINE VOLTAGE VAB?

$$f_{VAB} = f_{\text{Sin}}$$

For example,
TO INCREASE THE FREQUENCY OF VAB YOU NEED TO INCREASE THE "FREQUENCY OF THE SINUSOIDAL SIGNALS" & VICE-VERSA

HOW TO CONTROL MAGNITUDE OF LINE VOLTAGE VAB?

For example,
TO INCREASE THE VOLTAGE VAB YOU NEED TO INCREASE THE "AMPLITUDE OF THE SINUSOIDAL SIGNAL" & VICE-VERSA

TERMINNOLOGY 1:

$$\text{CARRIER RATIO} = f_t / f_s$$

f_t = frequency of triangular signal

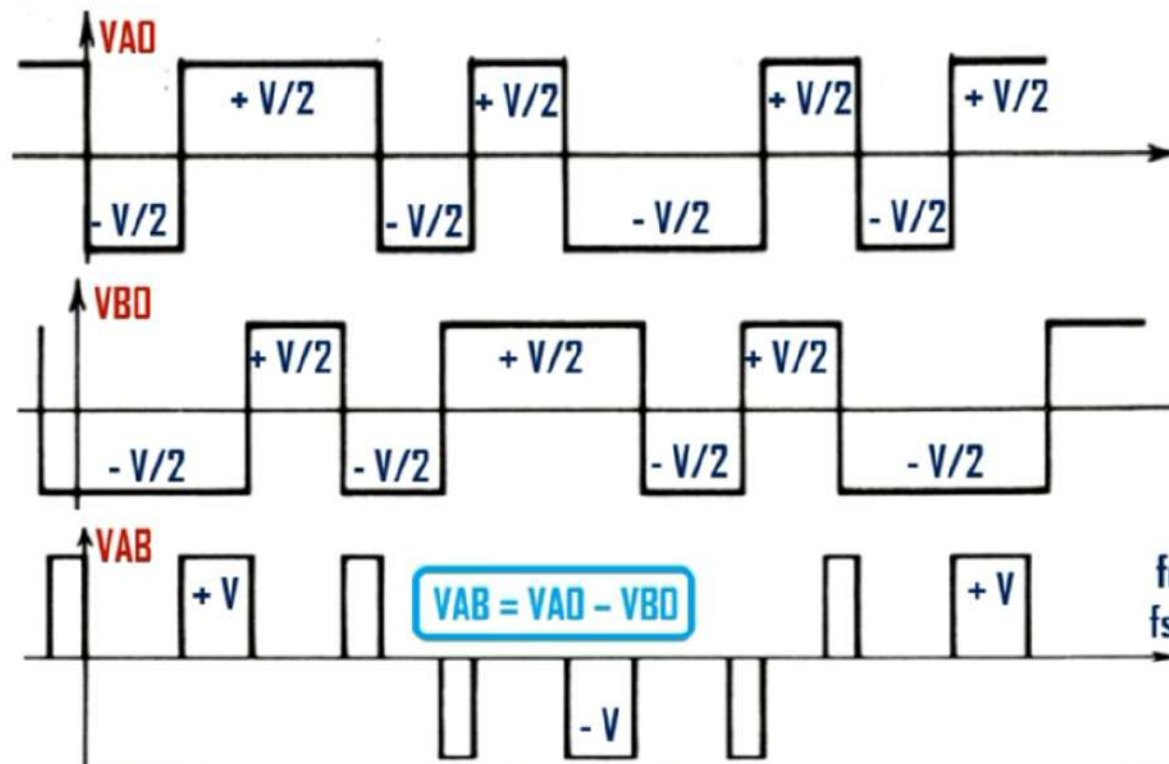
f_s = frequency of sinusoidal signals

TERMINNOLOGY 2:

$$\text{MODULATION INDEX} = A_s / A_t$$

A_m = Amplitude of sinusoidal signal

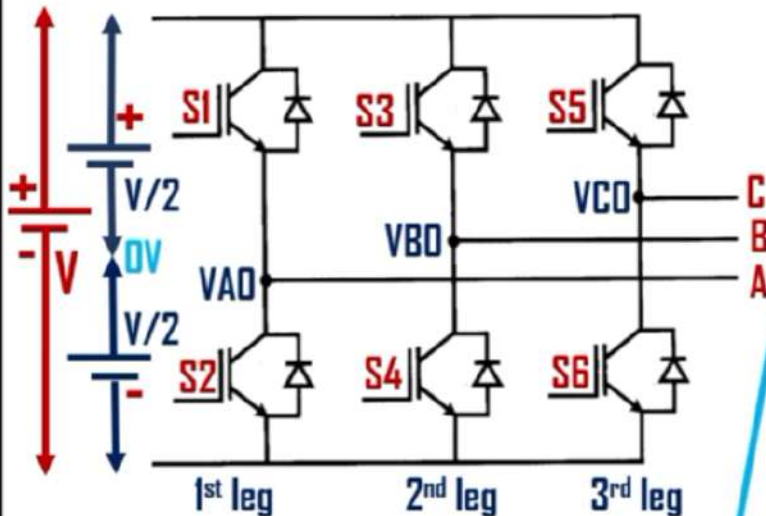
A_c = Amplitude of triangular signal



NOTE : Pulses in each half cycle have different widths (central pulse is wider). Less harmonics (Fourier analysis)

VOLTAGE SOURCE INVERTER FED 3 PHASE IM - SIX STEP SWITCHING

HOW FREQUENCY CONTROL IS DONE USING SIX STEP SWITCHING ?



V_{AO} , V_{BO} , V_{CO} = POLE VOLTAGES

(You can directly control these pole voltage by controlling the gate voltages of IGBT switches!)

For example:

- When **S1 (upper)** turned ON then, $V_{AO} \rightarrow +V/2$
- When **S2 (lower)** turned ON then, $V_{AO} \rightarrow -V/2$
- When **S3 (upper)** turned ON then, $V_{BO} \rightarrow +V/2$
- When **S4 (lower)** turned ON then, $V_{BO} \rightarrow -V/2$
- When **S5 (upper)** turned ON then, $V_{CO} \rightarrow +V/2$
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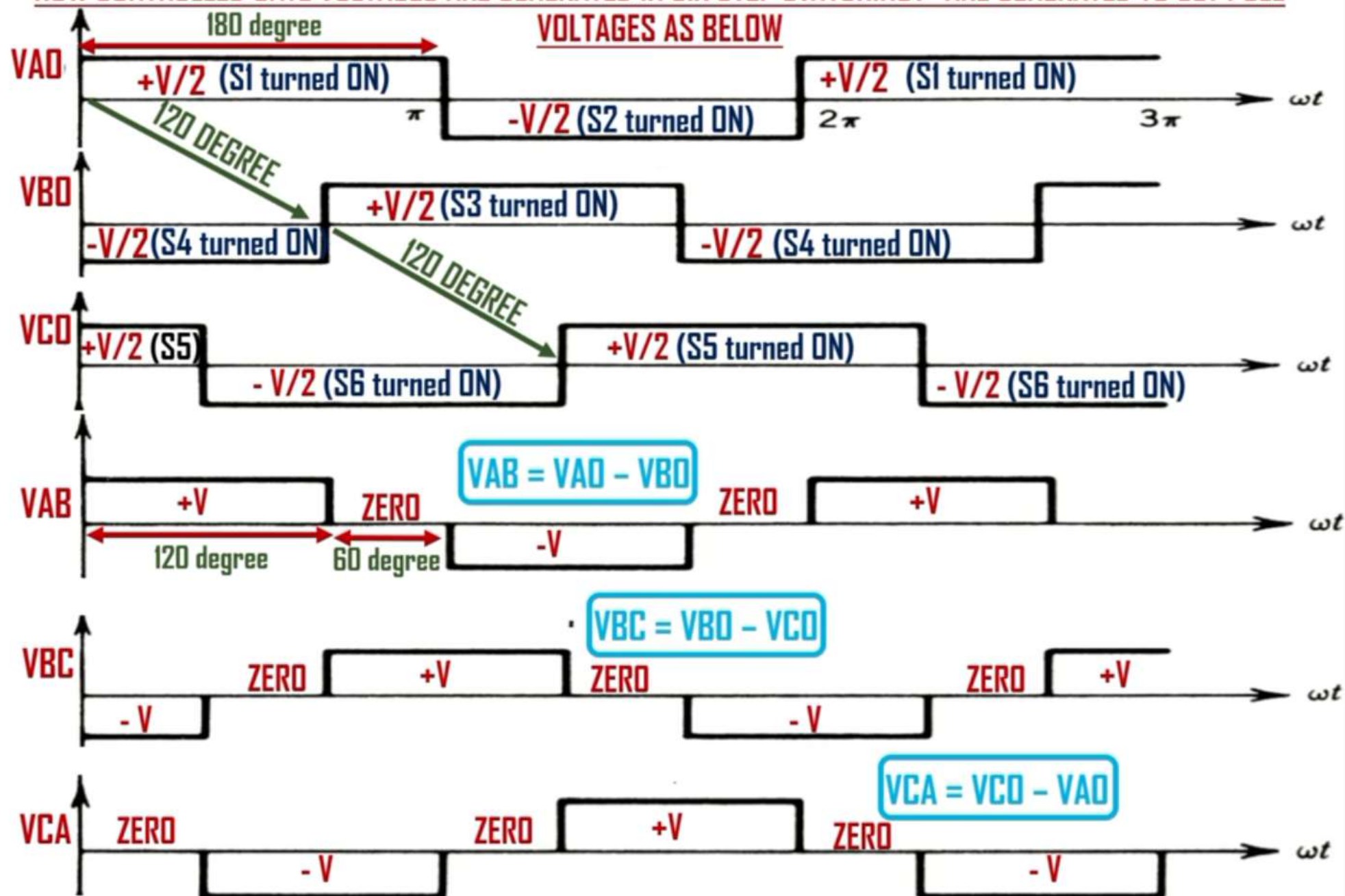
But in V/F control, what we need to control is "frequency" of line voltages (V_{AB} , V_{BC} , V_{CA})

If we control the pole voltages, then it is possible to control the frequency of line voltages

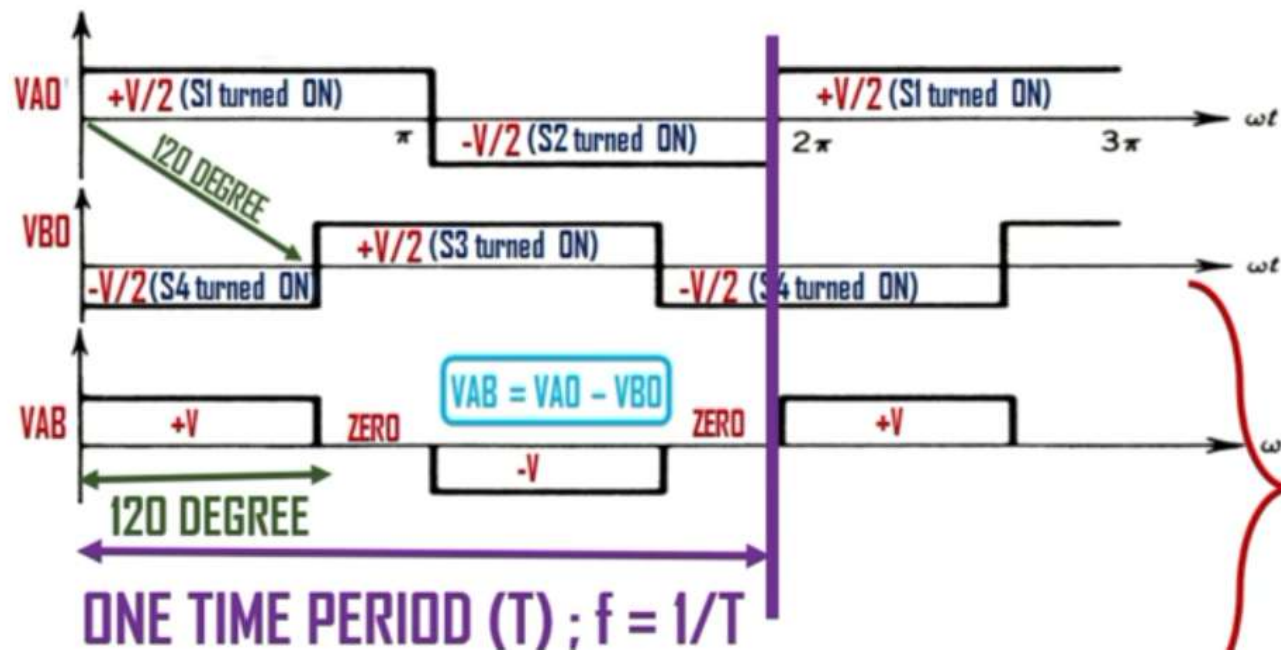
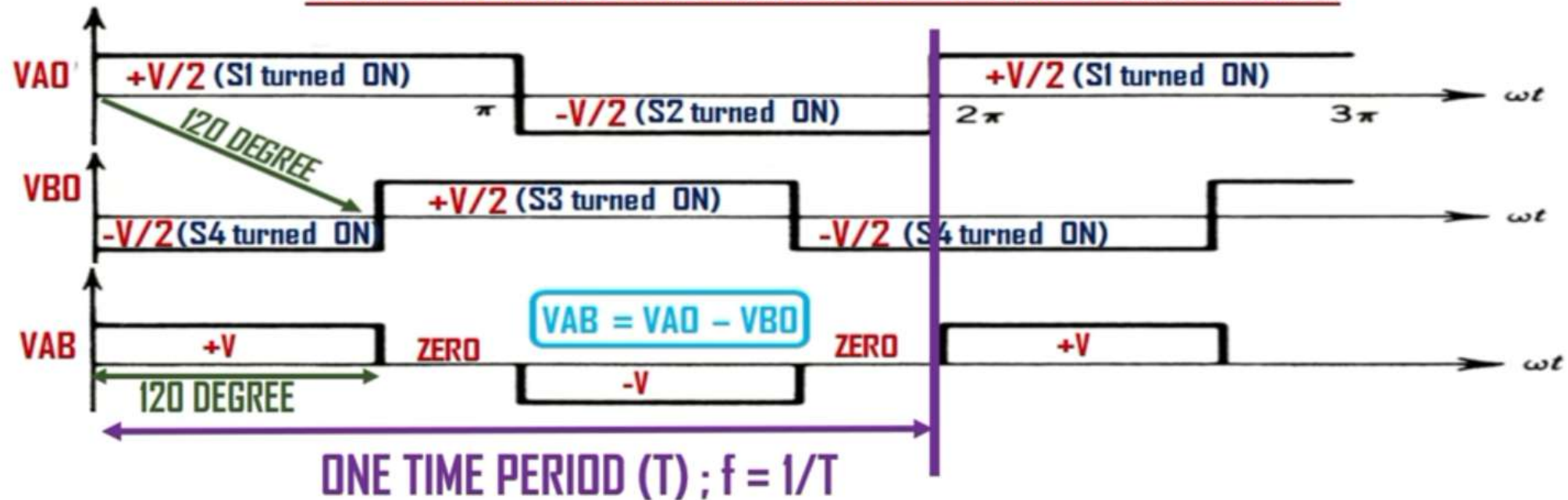
$$\begin{aligned} V_{AB} \text{ (0 DEG Phase shift)} &= V_{AO} - V_{BO} \\ V_{BC} \text{ (120 DEG Phase shift)} &= V_{BO} - V_{CO} \\ V_{CA} \text{ (240 DEG Phase shift)} &= V_{CO} - V_{AO} \end{aligned}$$

HOW CONTROLLED GATE VOLTAGES ARE GENERATED IN SIX STEP SWITCHING?- ARE GENERATED TO GET POLE

VOLTAGES AS BELOW



HOW TO CONTROL THE FREQUENCY OF LINE VOLTAGE VAB?



"Here Frequency of Line voltage VAB increases - By **increasing the switching frequency** of S1, S2, S3 & S4 OR in other words by **decreasing the switching time** of S1, S2, S3, S4"

HOW TO FIND THE PHASE VOLTAGE VAN ?

$$VAN = VAO - VNO \quad (1) \quad VBN = VBO - VNO \quad (2) \quad VCN = VCO - VNO \quad (3)$$

$$VNO = VAO - VAN \quad (4) \quad VNO = VBO - VBN \quad (5) \quad VNO = VCO - VCN \quad (6)$$

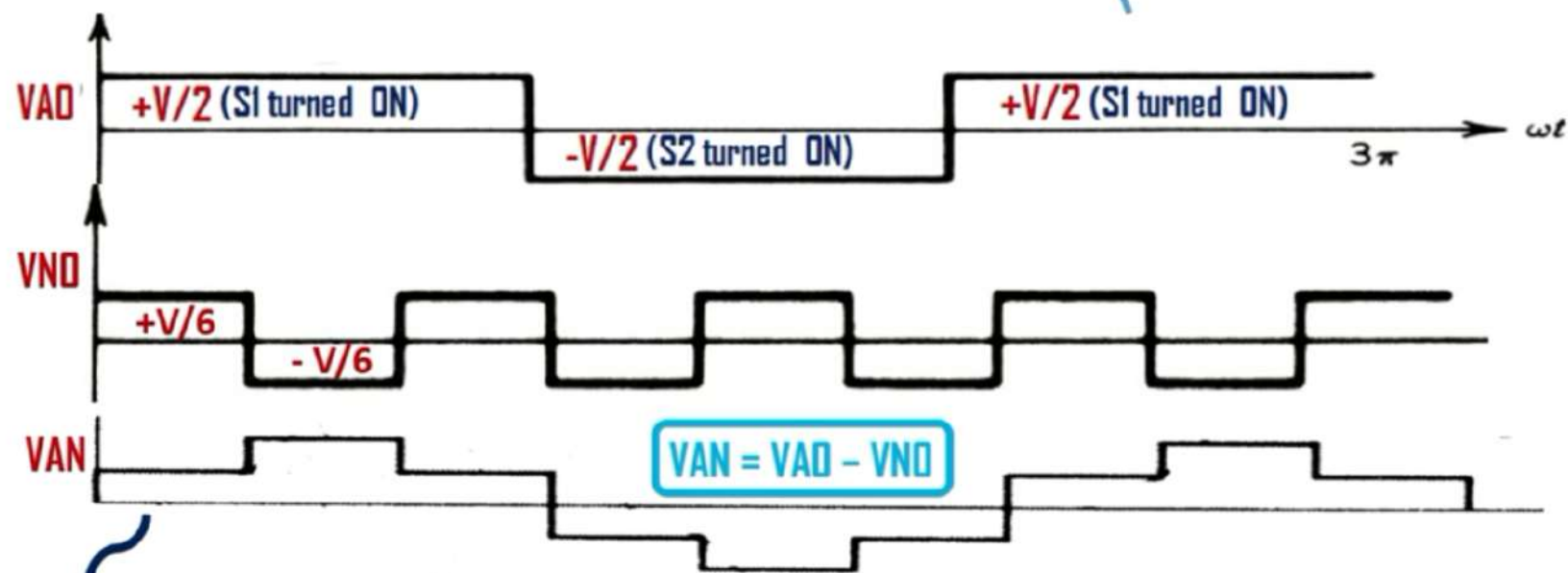
$$\text{ADDING 4, 5, 6} \rightarrow 3VNO = VAO + VBO + VCO - (VAN + VBN + VCN)$$

Substitute (7) in (1)

For a balanced three phase operation: $VAN + VBN + VCN = 0$

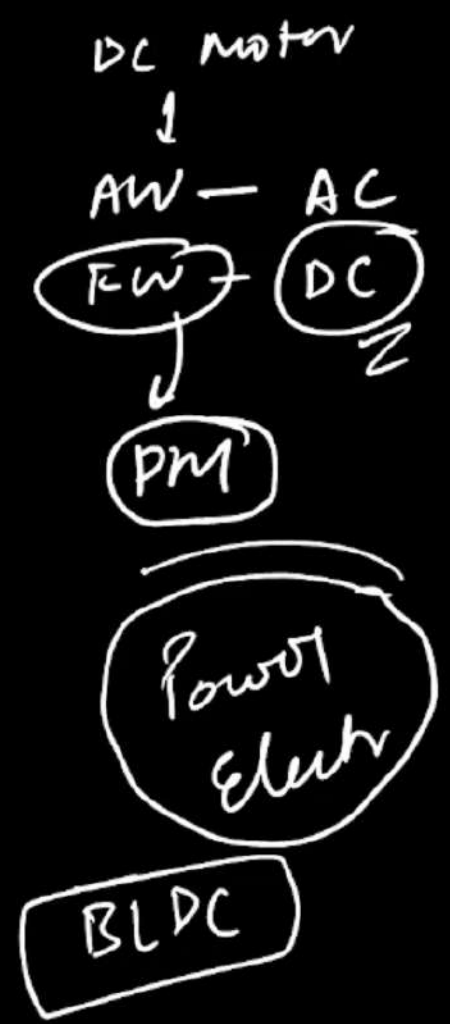
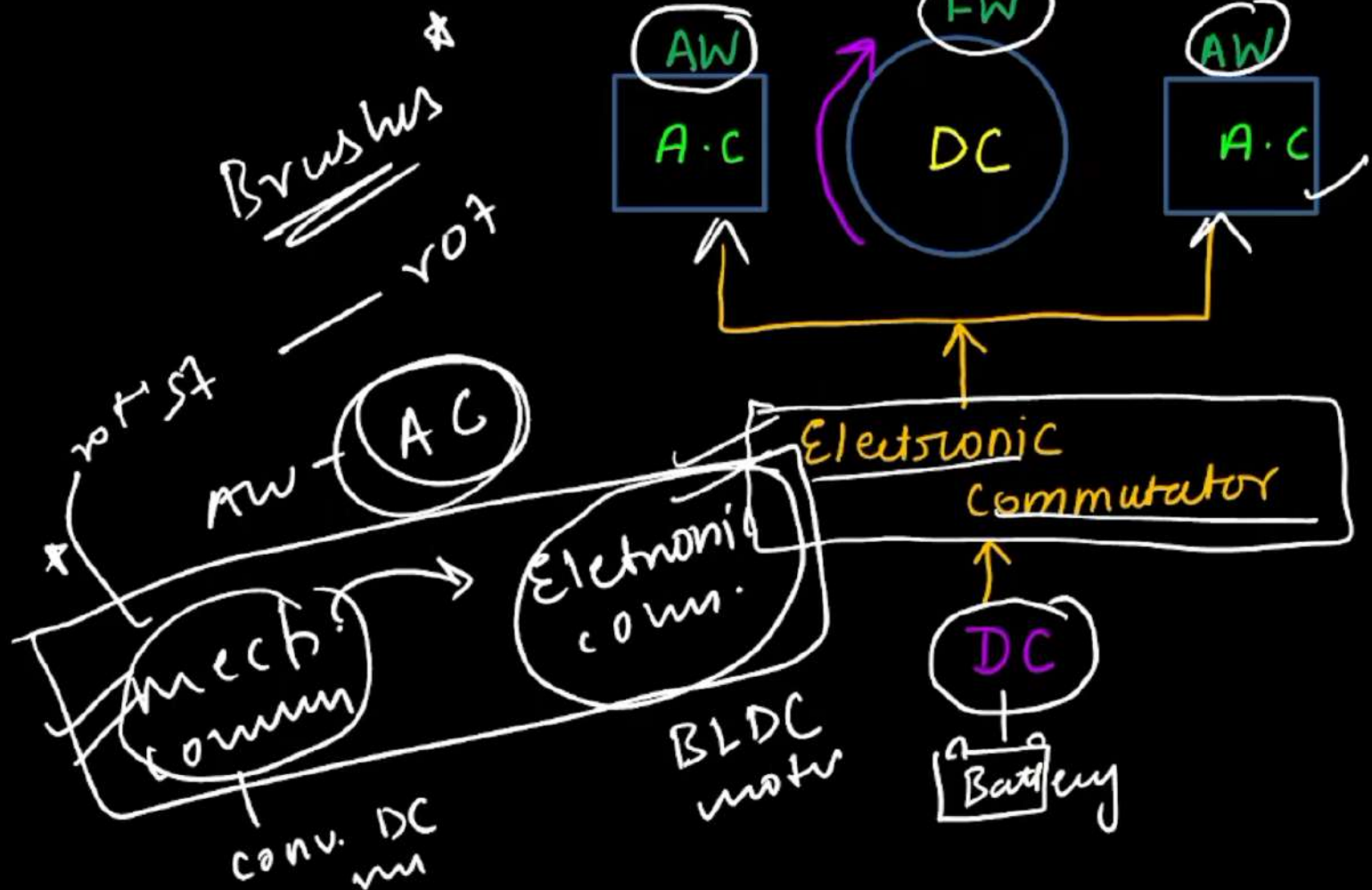
$$\text{So, } 3VNO = VAO + VBO + VCO \quad (\text{OR}) \quad VNO = (VAO + VBO + VCO)/3 \quad (7)$$

$$VAN = \frac{2}{3} (VAO) - \frac{1}{3} (VBO + VCO)$$

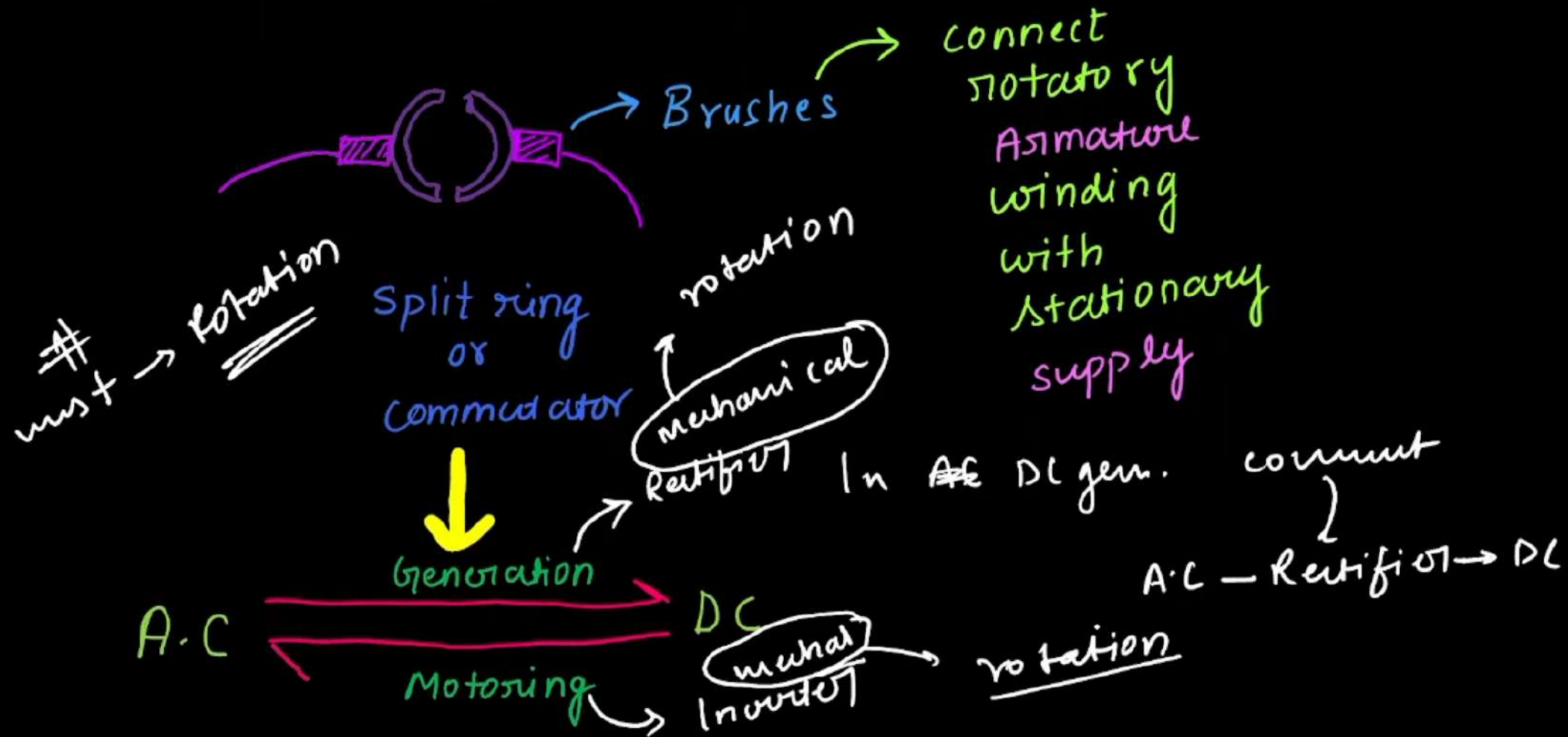


SIX STEPPED WAVE (PHASE VOLTAGES) $\rightarrow$ THE REASON FOR THE NAME SIX STEP SWITCHING

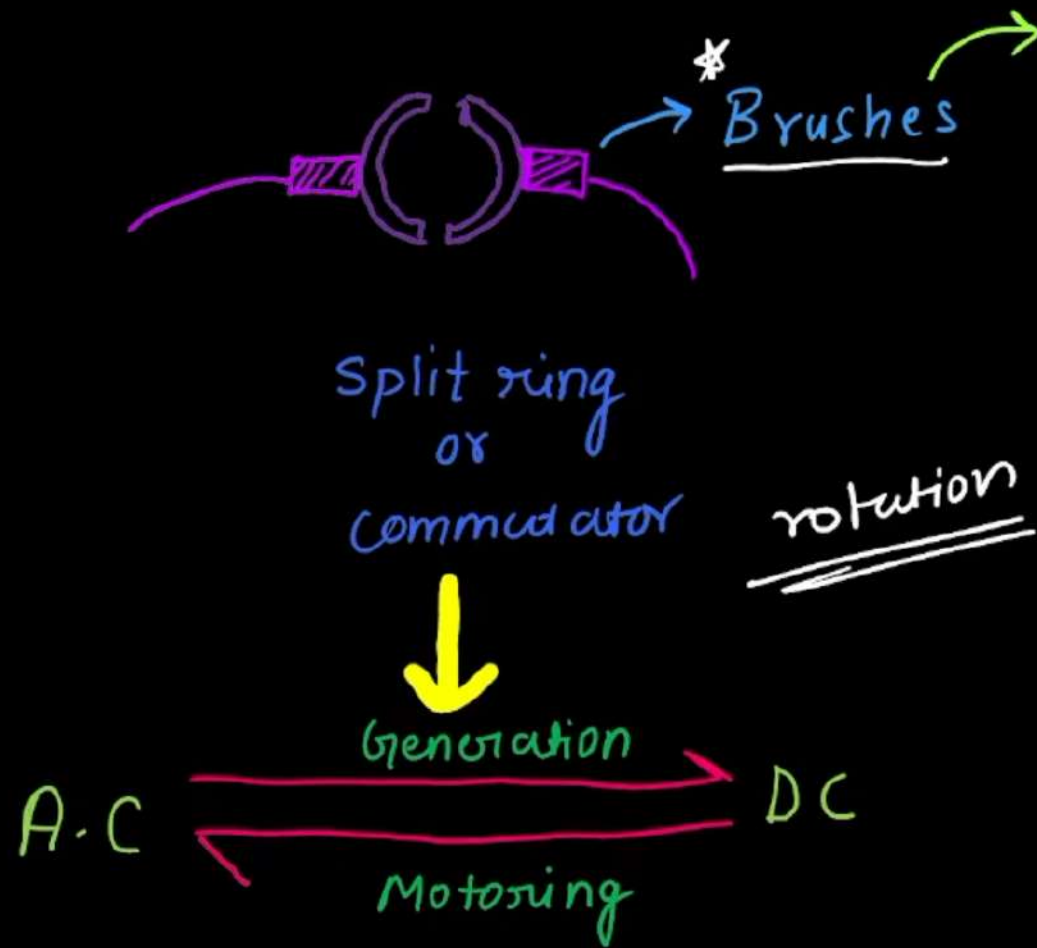
Brushless DC Motor *



Role of Commutator and Brushes



Role of Commutator and Brushes



connect
rotatory
Armature
winding
with
stationary
supply



rotating
commutator
↓
stationary
supply.

A.C in DC machine

Induced voltage " $\mathcal{E}$ " $\rightarrow$
{ Static or dynamic }
 ↓ ↓
variable flux flux cutting

In DC machine;
 Armature winding $\rightarrow$ A.C
 Field winding $\rightarrow$ D.C

Alternating (A.C)
in
nature.



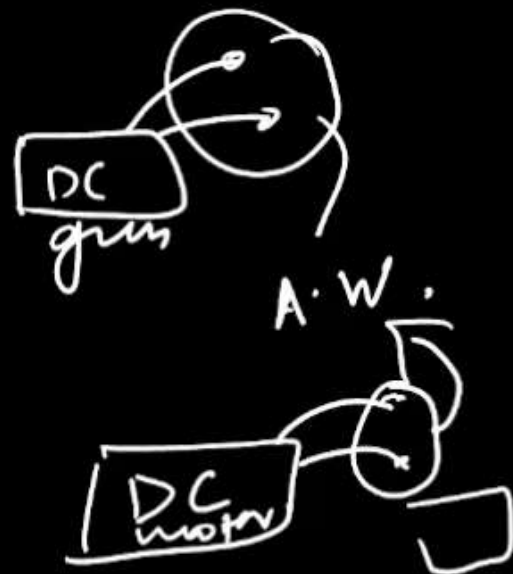
A.C in DC machine

Induced voltage " $\mathcal{E}$ "
{ Static or dynamic }
↓ variable flux ↓ flux cutting

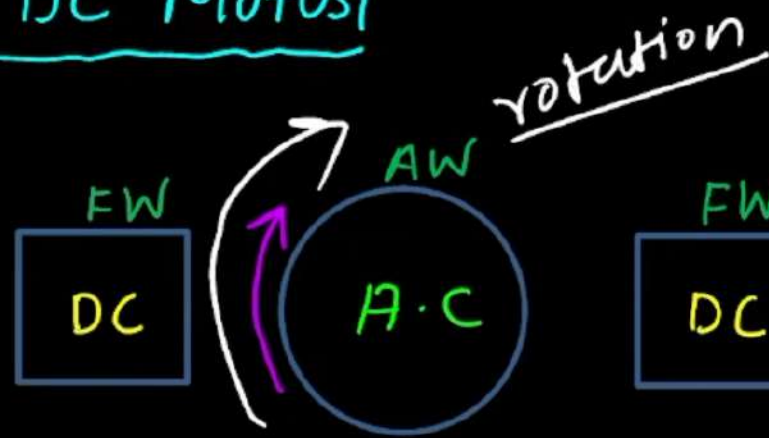
Alternating (A.C)
in
Nature.

In DC machine;

PMDC
PM.
Armature winding → A.C
Field winding → D.C
const. polarity flux



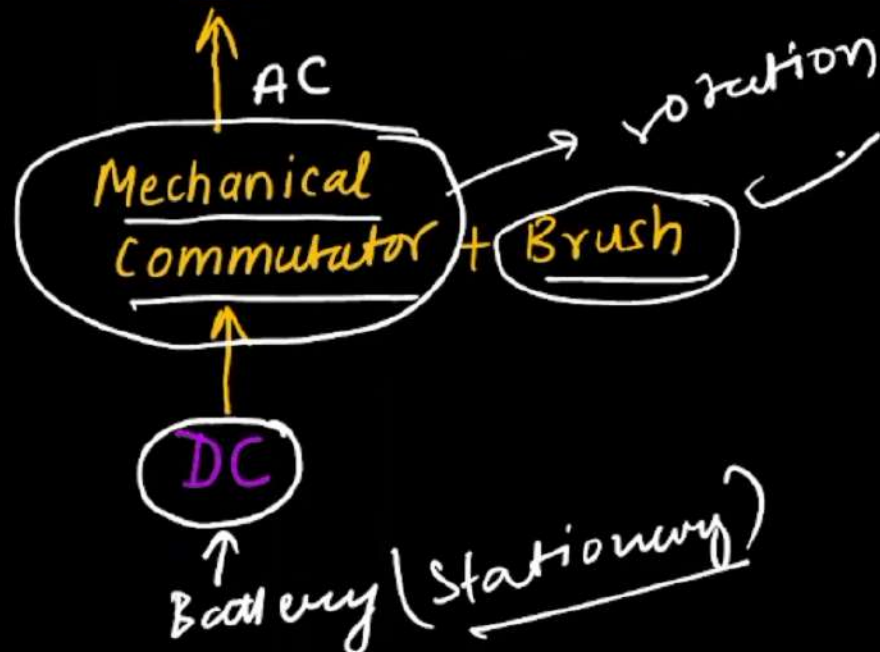
Conventional DC Motors



← $\begin{cases} \text{AW} \rightarrow \text{AC} \\ \text{FW} \rightarrow \text{DC} \end{cases}$

supply - DC

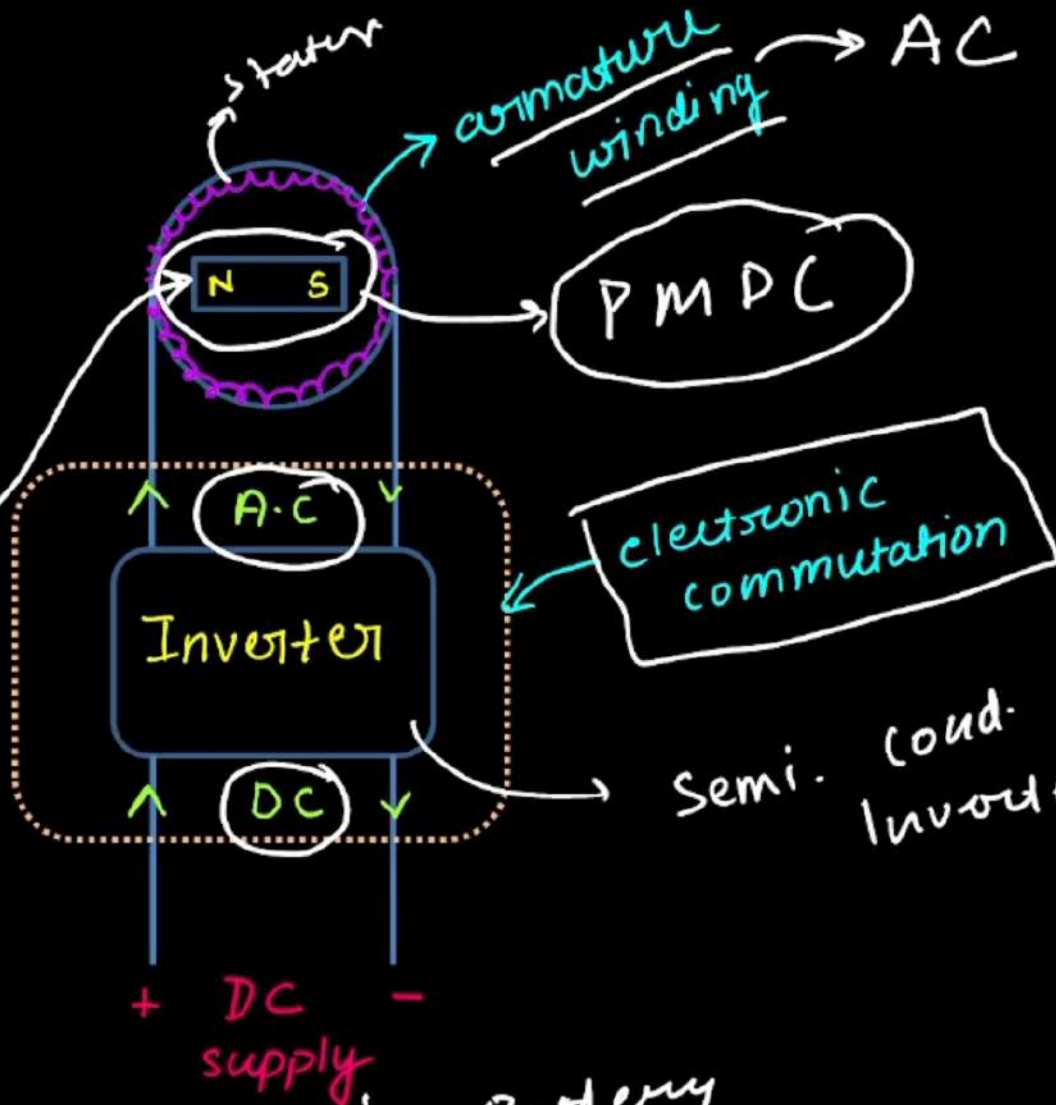
AW - stator



#BLDC motor

Brushless
IPMDC

DC
F.W
PM
Rotor



fix FW

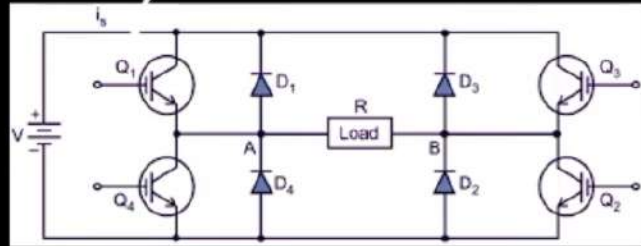
Semi. cond. based
Inverter.

Performance



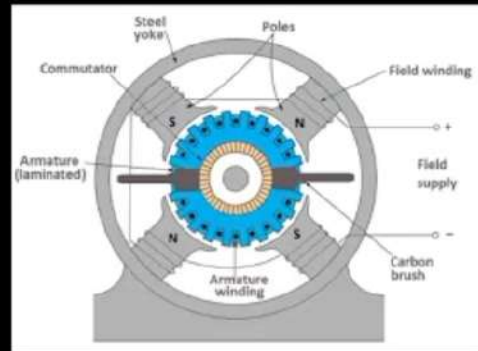
BLDC
motor

+



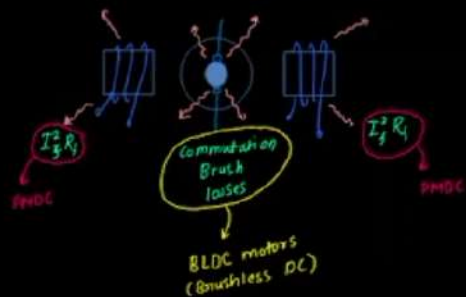
Auto-synchronized
Inverter

=



Conventional
DC motor

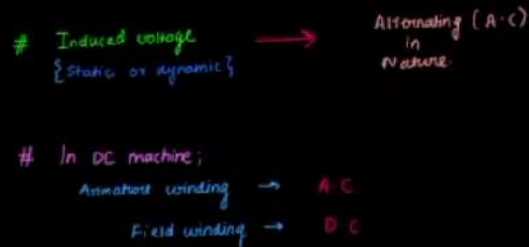
Conventional DC motor



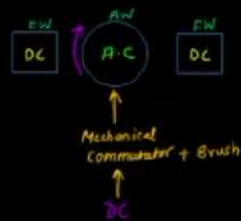
Role of Commutator and Brushes



A.C in DC machine



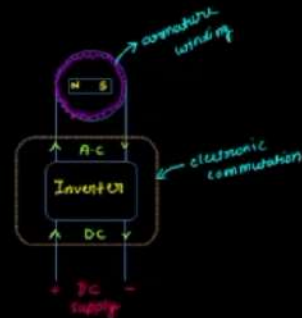
Conventional DC Motor



Brushless DC Motor



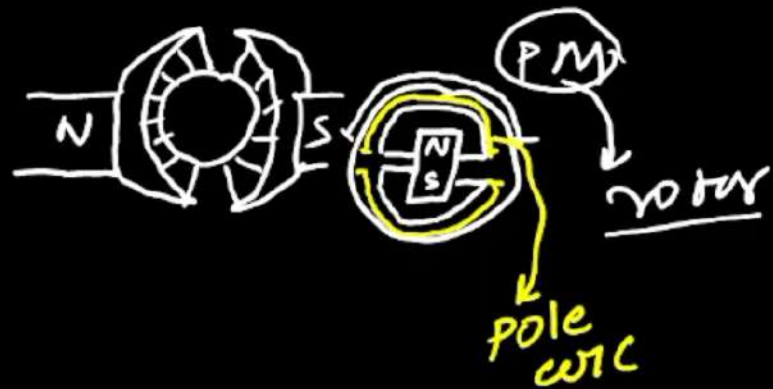
BLDC motor



TYPES OF BLDC



Classification of BLDC motors

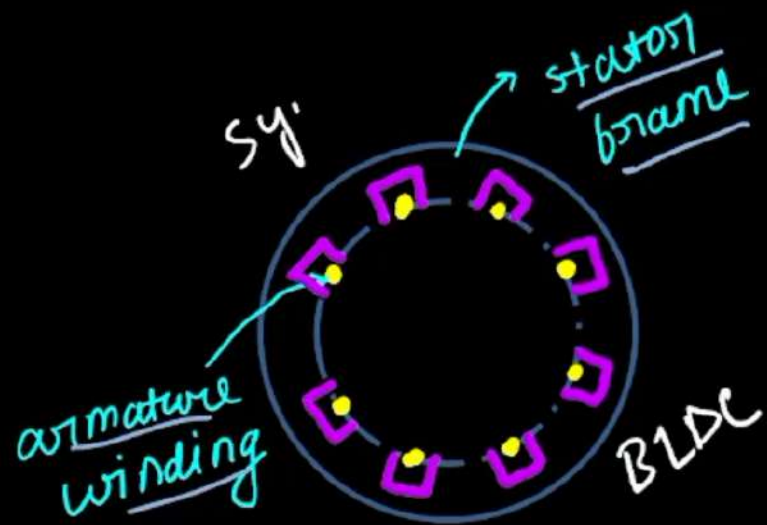


On the basis of
motor
construction

On the basis
of length
of poles arc

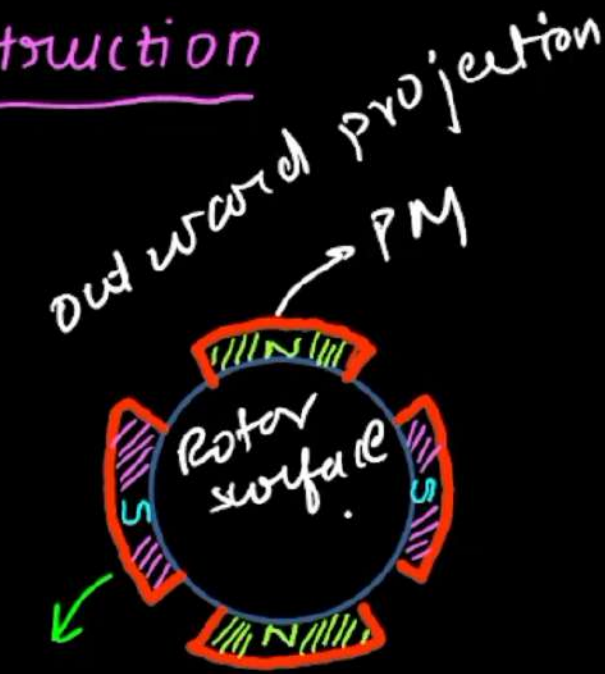
on the basis of
electronic
commutator

On the basis of rotor construction



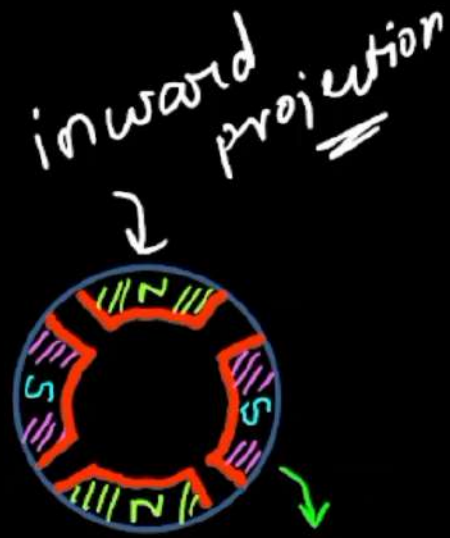
Stator Frame

Similar to IM stator

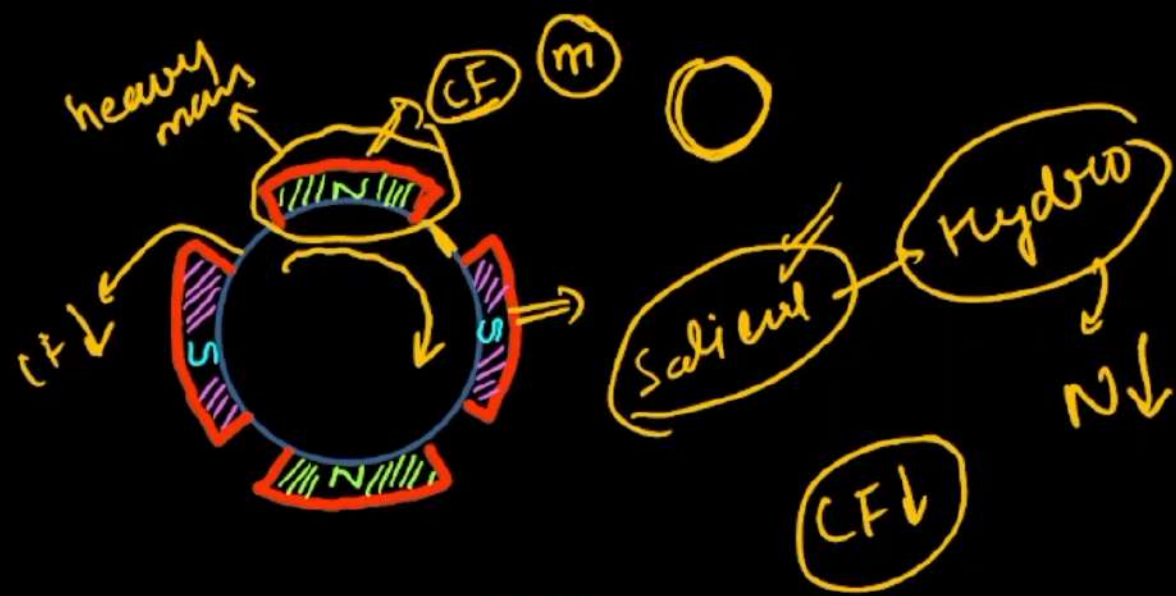


Surface mount
PM
rotor

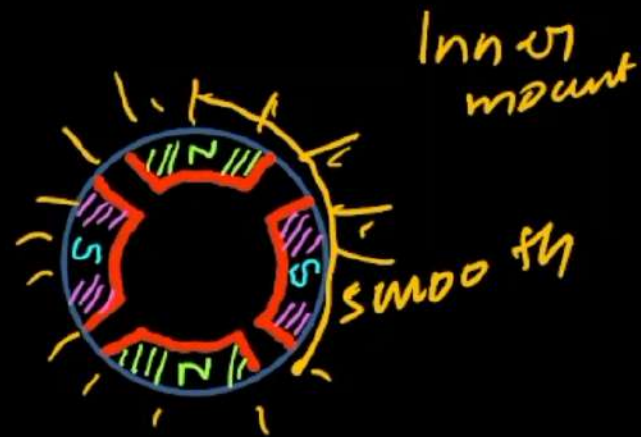
Rotor structure



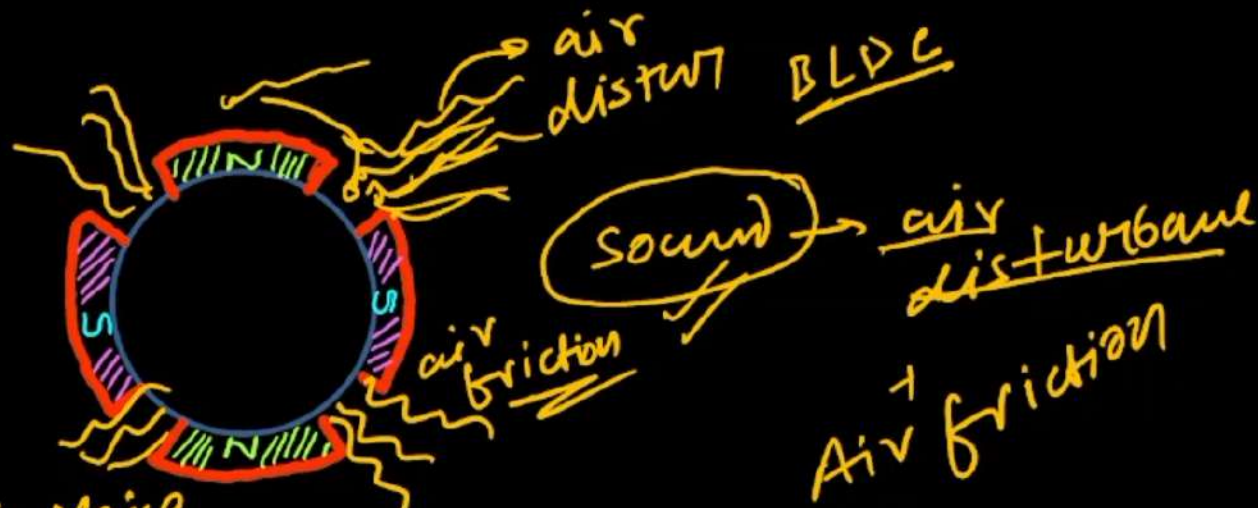
Interior mount
PM
rotor



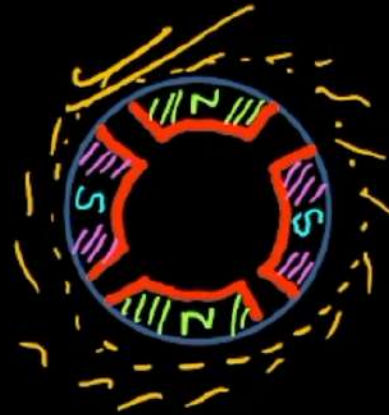
- * Uneven surface
- * Magnetic + surface saliency
- * Speed limitation due to centrifugal force



- * Smooth surface
- * Magnetic saliency
- * High speed operation

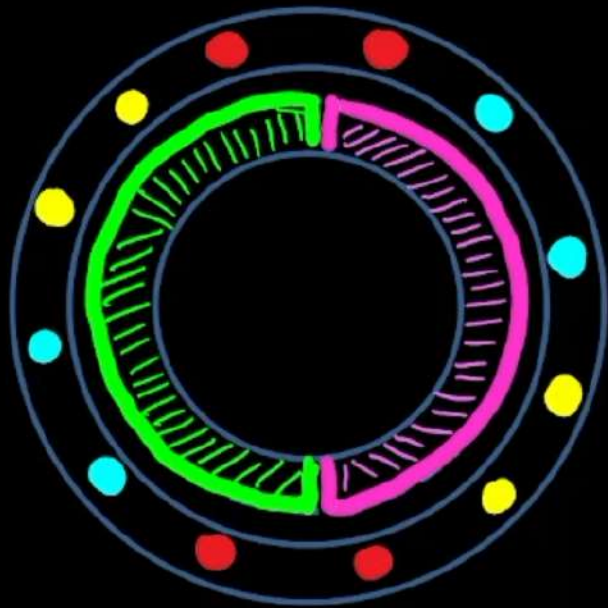


- * Noise
- * Uneven surface
- * Magnetic + Surface saliency
- * Speed limitation due to centrifugal force

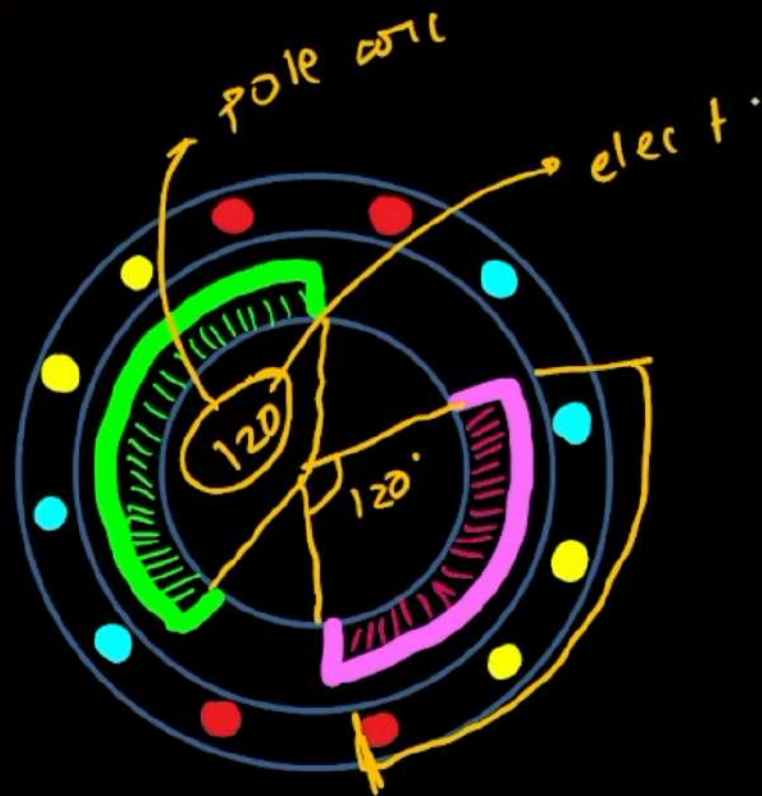


- * Smooth Surface
- * Magnetic saliency
- * High speed operation
- * Noiseless operation

On the basis of length of pole arc



180° Pole Arc.

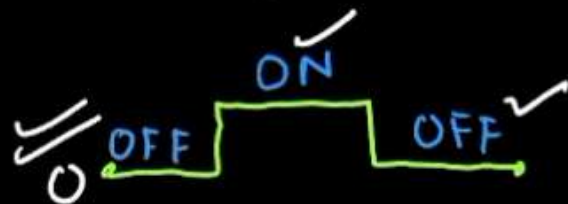


120° pole arc

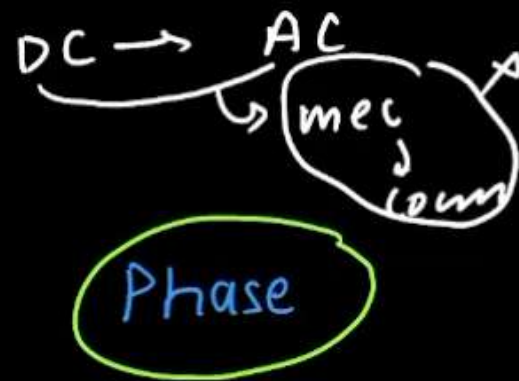
Role of pulses and Phases

Pulse

* Used to turn ON and OFF the switches at proper intervals.

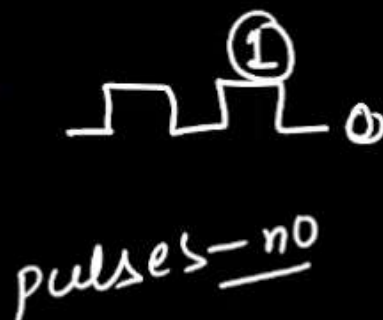


* Decides the no. of switches in electronic commutation ckt.



Phase

* Decides no. of windings phase in the armature of BLDC



Role of pulses and Phases

Pulse

* Used to turn ON and OFF the switches at proper intervals.



* Decides the no. of switches in electronic commutation ckt.



Phase

* Decides no. of windings phase in the armature of

BLDC

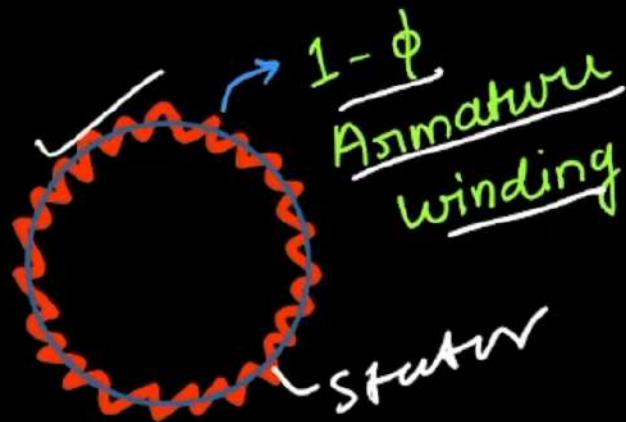


Decoding

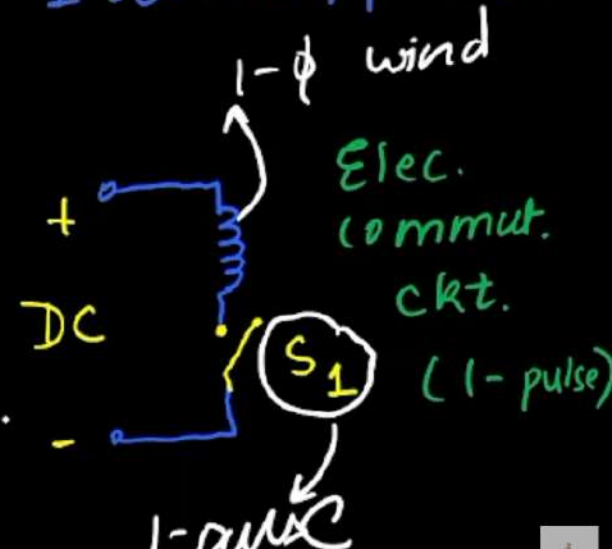
$1-\phi$ 1 -pulse $\rightarrow$ BLDC have $1-\phi$ winding on stator.

Code $\rightarrow$ Elec. comm. ckt use single pulse to ON and OFF the single switch of $1-\phi$ winding.

(1 switch/phase)



ON
OFF

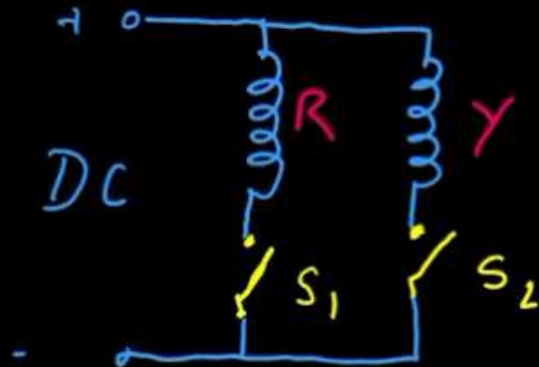
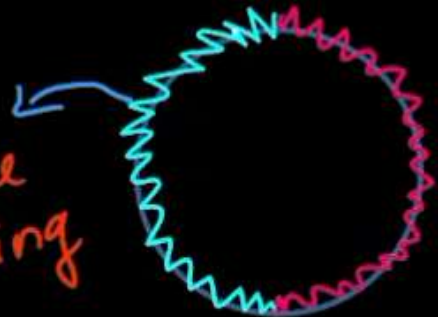


2- ϕ $\rightarrow$ stator have 2 phase winding

2-Pulse

$\rightarrow$ Elec. comm. ckt. use two pulses to ON and OFF the 2 switches of 2- ϕ winding (1 switch / phase)

2 ϕ
Armature
winding

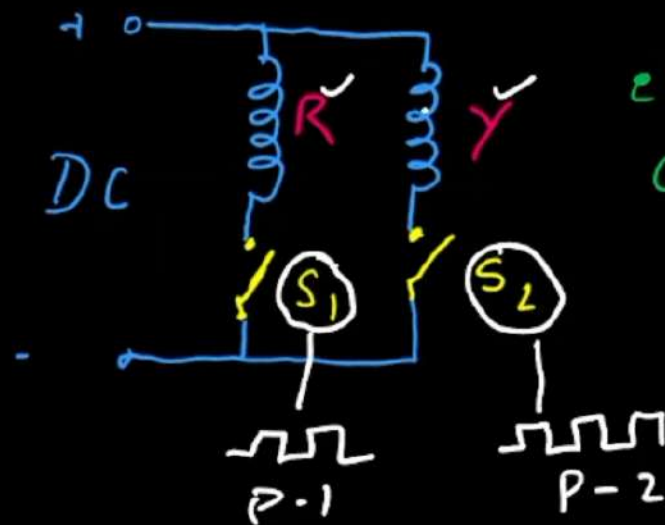
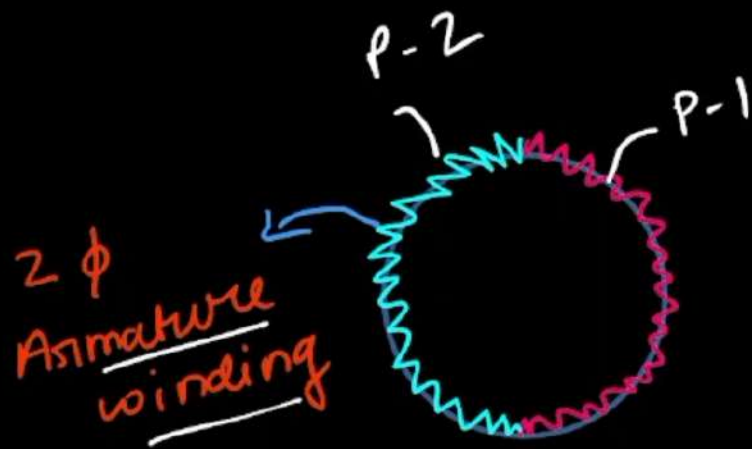


electronic
commutation
ckt
(2-pulse)

2- ϕ 2-Pulse $\rightarrow$ stator have 2 phase winding

$$\frac{\text{No. of}}{\text{1 u}} = \frac{\text{No. of}}{\text{pulse}}$$

$\rightarrow$ Elec. comm. ckt. use two pulses to ON and OFF the 2 switches of 2- ϕ winding (1 switch / phase)

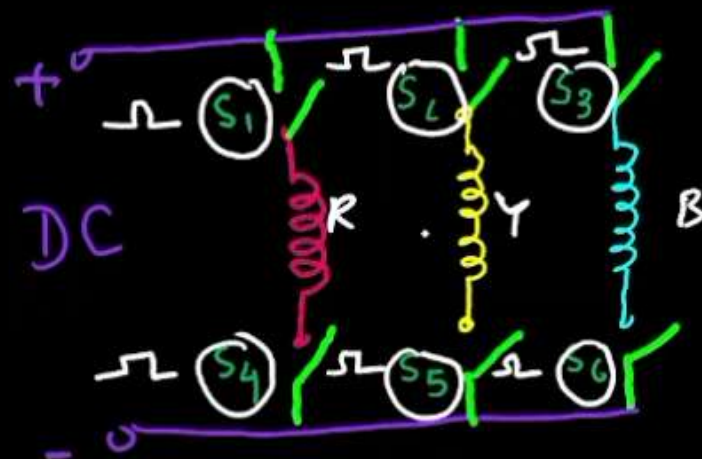
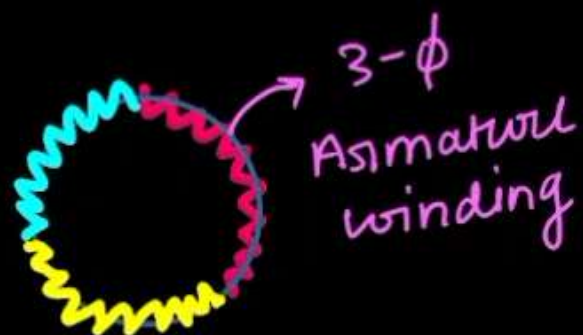


electronic
commutation
ckt
(2-pulse)

Mostly used Configuration

* $3-\phi$ → BLDC stator have $3-\phi$ winding

* 6-Pulse → Elec. comm. ckt use 6 pulses to ON and OFF the 6 switches of $3-\phi$ winding (2 switch/phase)



Elec. comm. ckt (6-pulse)

WORKING PRINCIPLE



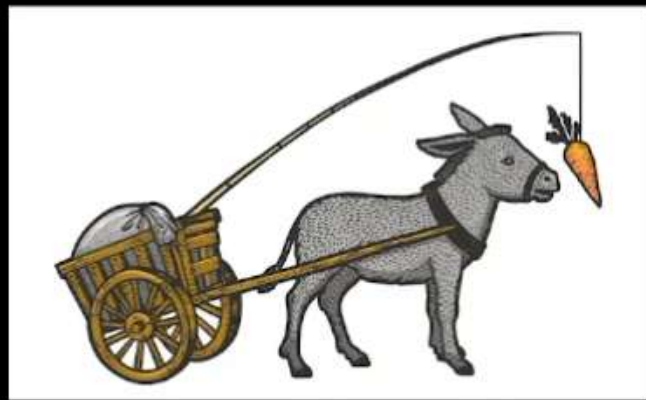
STORY



+



TRICK

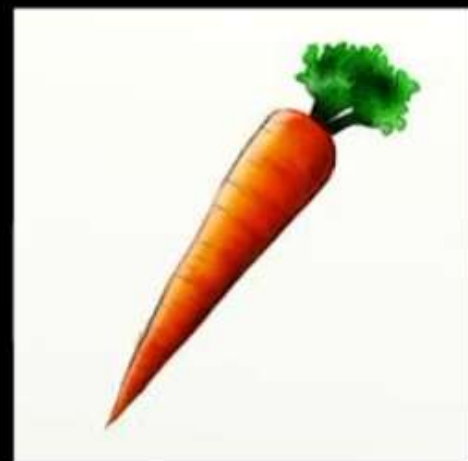




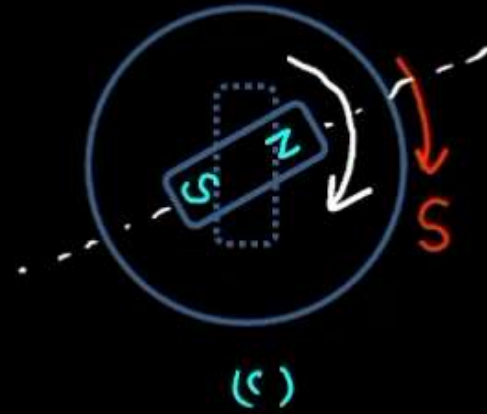
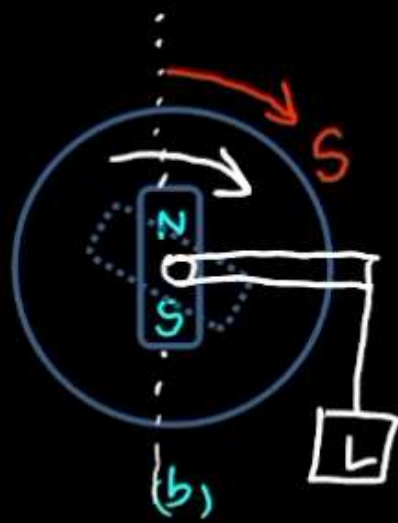
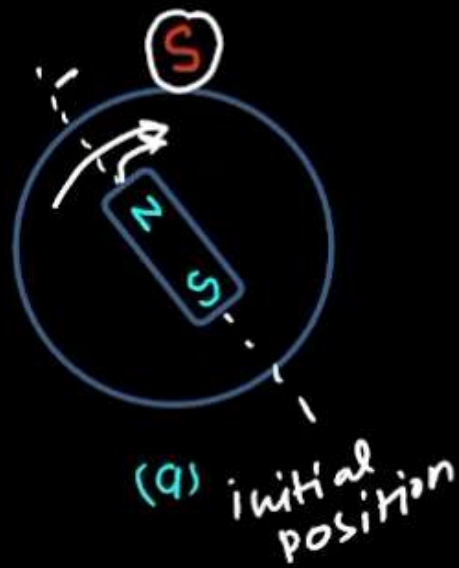
Permanent
Magnet
of
BLDC motor



Mechanical
Load



Magnetic
field



⚡
we need
"Rotating Magnetic Field"

Principle of operation

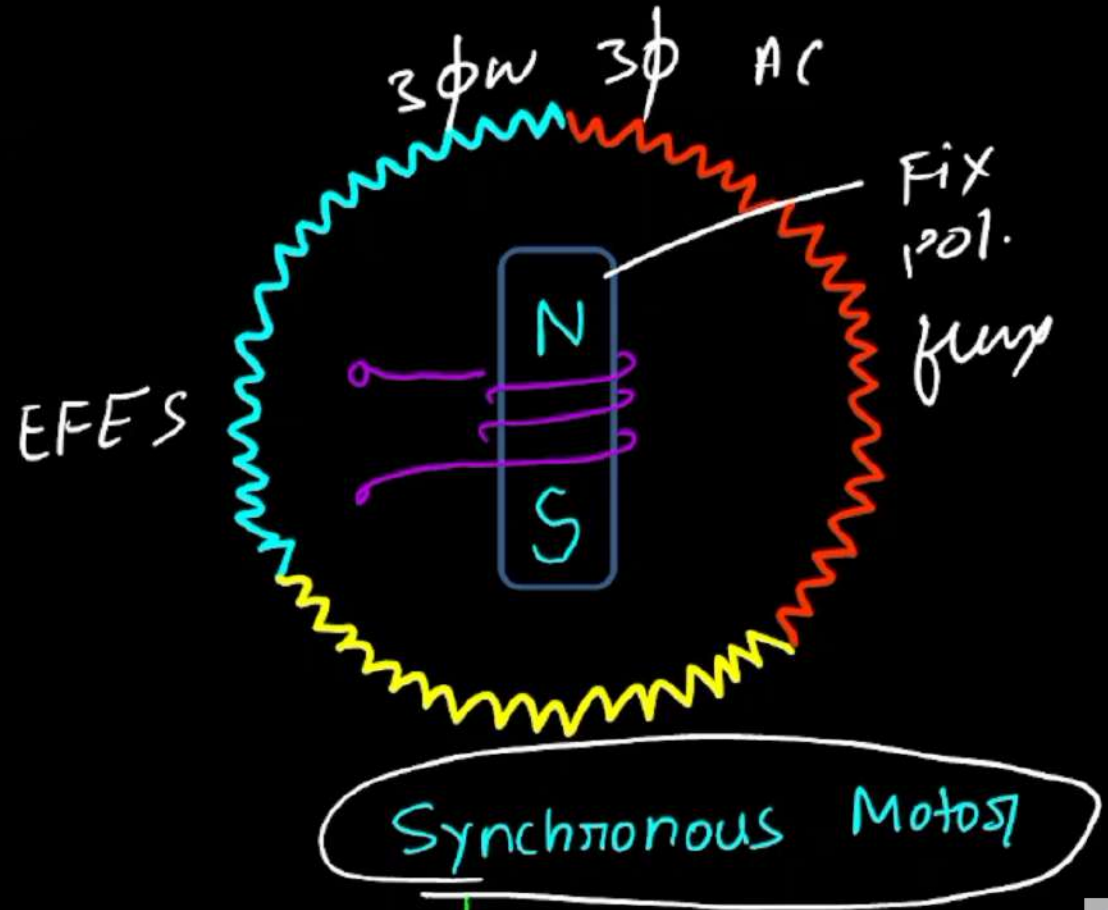
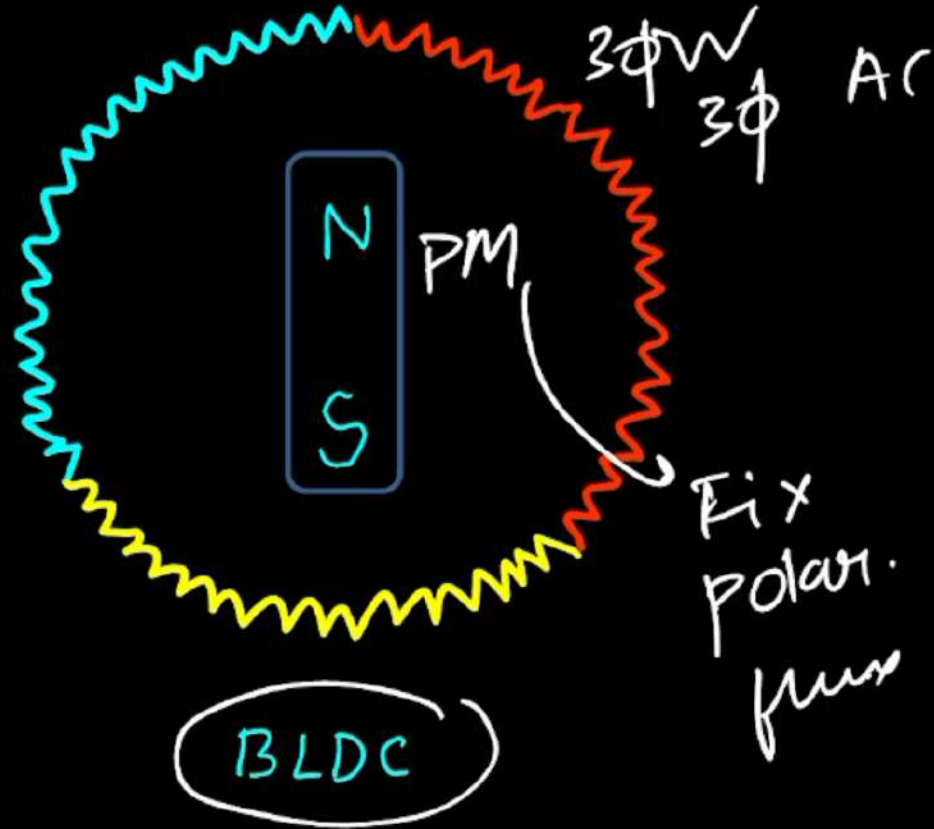
(RMF)

* Generate rotating magnetic field via 3- ϕ armature winding.

* Attract permanent magnetic rotor via rotating magnetic field.

* Motor start rotating

Problem of starter



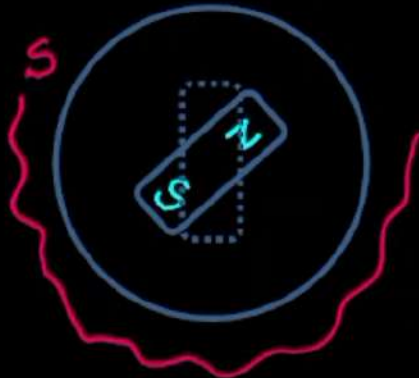
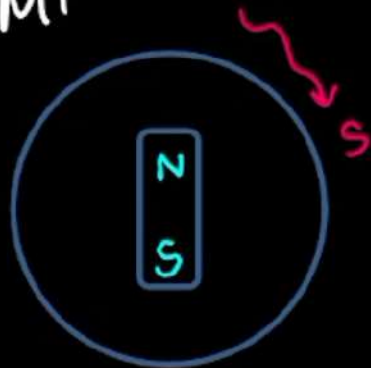
problem of self
starting

Self-start problem in synchronous motor

- ① Inertia $\uparrow \uparrow$
- ② RMF speed $\uparrow$

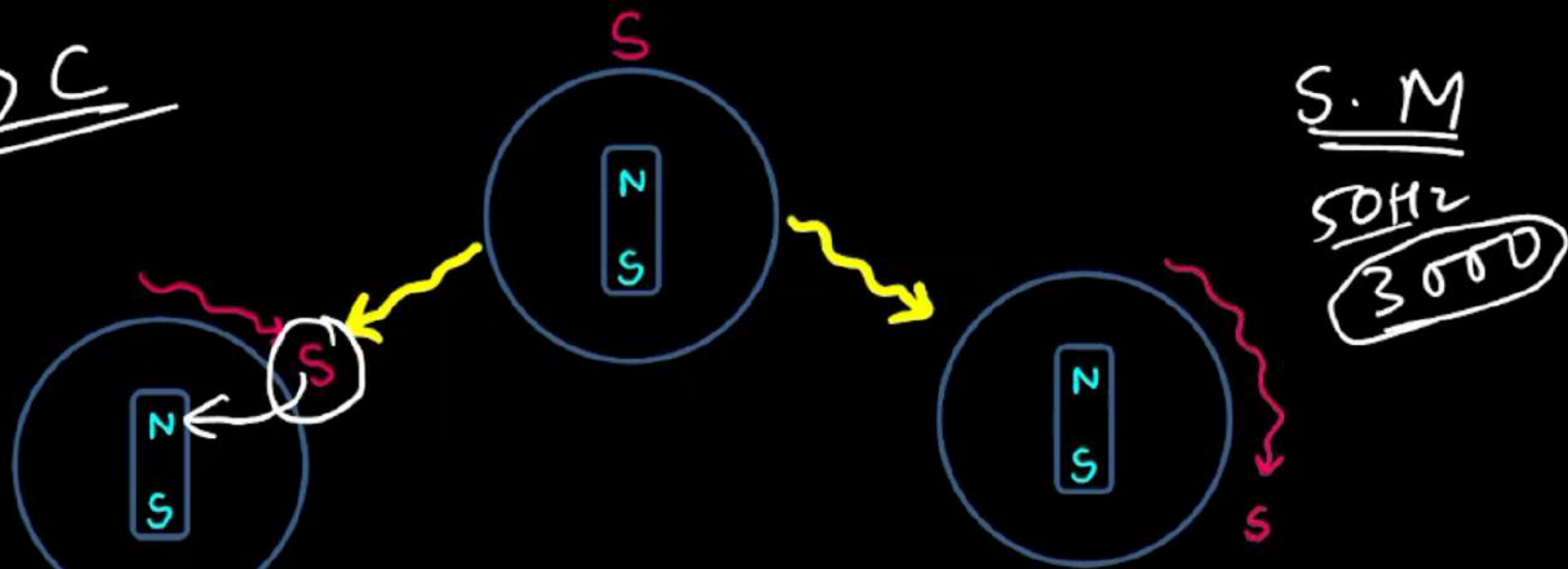
$$N_s = \frac{120f}{P} = \frac{120 \times 50}{2} = 3000 \text{ RPM}$$

< -5



"Rotor will only oscillate"

BLDC



S.M
50Hz
3000

* RMF have full information about Rotor position

* Control the speed of RMF
as per rotor position

* Magnetic Locking

at start

SELF START

* RMF has No information about Rotor position

* No control over the speed of RMF as per rotor position

* No magnetic Locking

at start

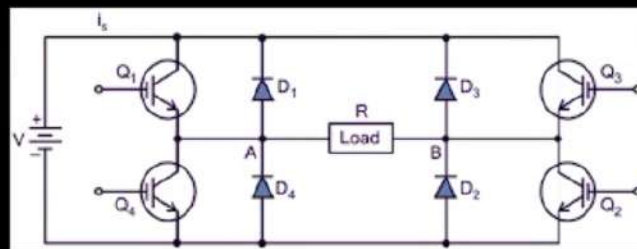
FAIL

Self start BLDC Requirement

* Positional sensor based 3- ϕ system

BLDC self start
rotor info
positional sensor

Able to control the speed of RMF
as per rotor position



Auto-synchronized
Inverter

Self start BLDC Requirement

posi

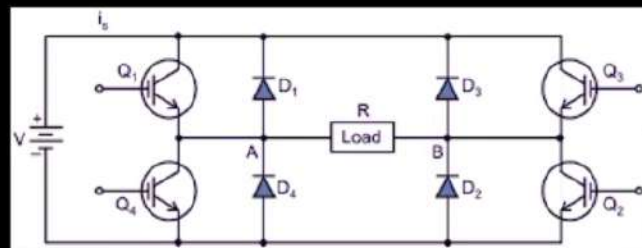
3 ϕ

* Positional sensor, based 3- ϕ system

BLDC self start
rotor info
positional sensor

Able to control the speed of RMF
as per rotor position

$$N_s = \frac{120f}{P}$$



Auto-synchronized
Inverter

Auto syn
electro

A simplified equivalent circuit of one phase is shown in Figure 6.48, where V_t is the voltage of the power supply, R_s is the resistance of the winding, L_s is the leakage inductance ($L_s = L - M$, where L is the self-inductance of the winding and M is the mutual inductance), and E_s is the back EMF induced in the winding by the rotating rotor.

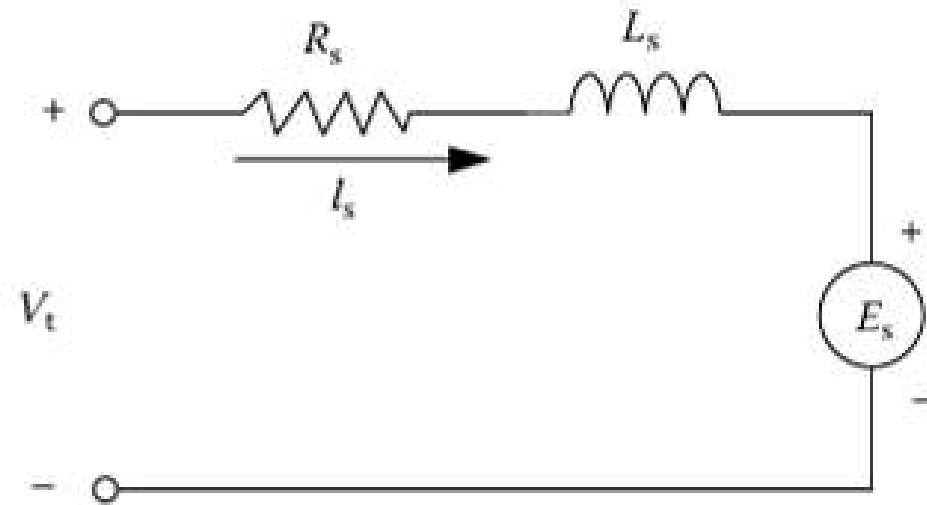


Figure 6: Simplified equivalent circuit of the BLDC motor.

Based on the equivalent circuit of Figure 6.48, the performance of the BLDC motor can be described by

$$V_t = R_s I_s + L_s \frac{dI_s}{dt} + E_s,$$

$$E_s = k_E \omega_r$$

$$T_e = k_T I_s$$

$$T_e = T_L + J \frac{d\omega_r}{dt} + B\omega_r$$

where k_E is the back EMF constant, which is associated with the PMs and rotor structure, ω_r is the angular velocity of the rotor, k_T is the torque constant, T_L is the load torque, and B is the viscous resistance coefficient. For steady-state operation, Equations can be simply reduced to

$$T_e = \frac{(V_t - k_E \omega_r) k_T}{R_s}$$

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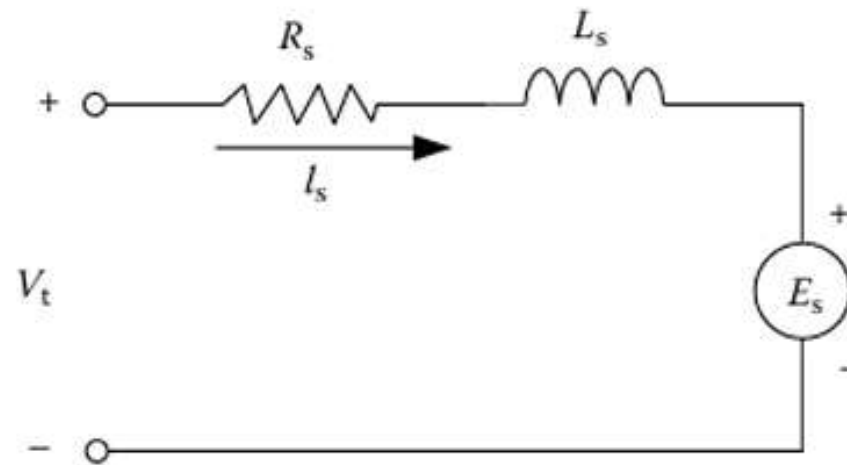


Figure 6: Simplified equivalent circuit of the BLDC motor.

Based on the equivalent circuit of Figure 6.48, the performance of the BLDC motor can be described by

$$V_t = R_s I_s + L_s \frac{dI_s}{dt} + E_s,$$

$$E_s = k_E \omega_r$$

~~★~~ $k_e = k_t = k_a \phi_p$

$$T_e = k_T I_s$$

$$T_e = T_L + J \frac{d\omega_r}{dt} + B \omega_r$$

where k_E is the back EMF constant, which is associated with the PMs and rotor structure, ω_r is the angular velocity of the rotor, k_T is the torque constant, T_L is the load torque, and B is the viscous resistance coefficient. For steady-state operation, Equations can be simply reduced to

$$T_e = \frac{(V_t - k_E \omega_r) k_T}{R_s}$$

The speed-torque performance with constant voltage supply is shown in Figure 7.

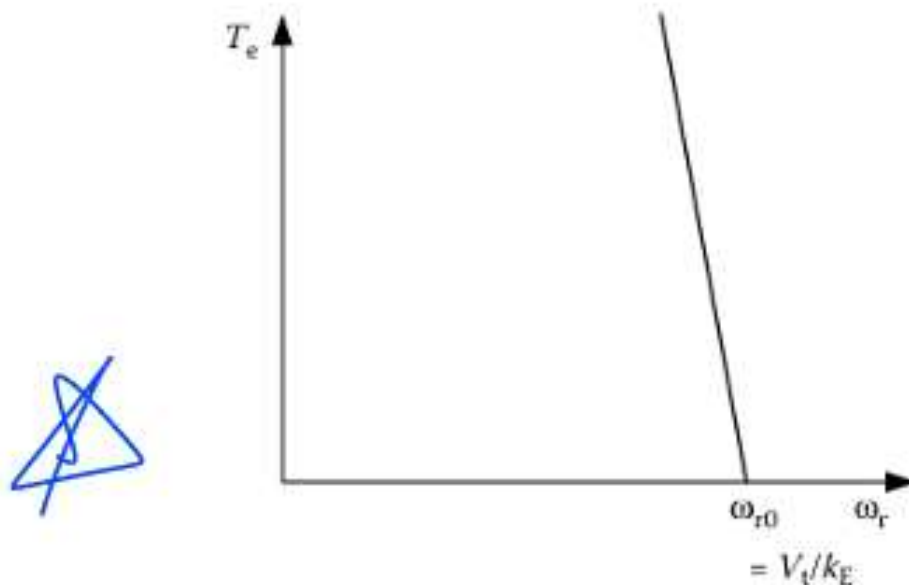


Figure 7: Speed-torque curve in the steady state with constant voltage.

It can be seen from Equation and Figure that at low speed, especially while starting, very high torque is produced, which results in very high current due to the low back EMF. This very high current would damage the stator windings.

With variable voltage supply, the winding current can be restricted to its maximum by actively controlling the voltage; thus, a maximum constant torque can be produced as shown in Figure

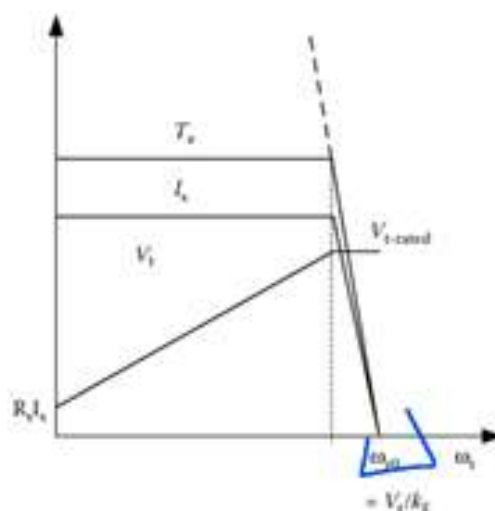


Figure 8: Speed-torque curve in the steady state with variable voltage supply.

For dynamic or transient operation, the performance of the BLDC machine is described by Equations. However, Laplace transform is helpful to simplify the analysis. Equations can be expressed by their Laplace forms as

$$V_t(s) = E_s(s) + (R_s + sL_s)I_s(s)$$

$$E_s(s) = k_E \omega_r(s)$$

$$T_e(s) = k_T I_s(s)$$

$$T_e(s) = T_L(s) + (B + sJ)\omega_r(s)$$

Thus, the transfer function of the BLDC motor drive system is

$$\omega_r(s) = \frac{k_T}{(R_s + sL_s)(sJ + B) + k_T k_E} V_t(s) - \frac{R_s + sL_s}{(R_s + sL_s)(sJ + B) + k_T k_E} T_L(s).$$

L_s and J in Equation represent the electrical and mechanical delay in transient operation. L_s determines how quickly the armature current builds up in response to a step change in the terminal voltage, where the rotor speed is assumed to be constant. J determines how quickly the speed builds up in response to a step change in the terminal voltage.

OUR MAIN AIM HERE
IS TO CONTROL
TORQUE

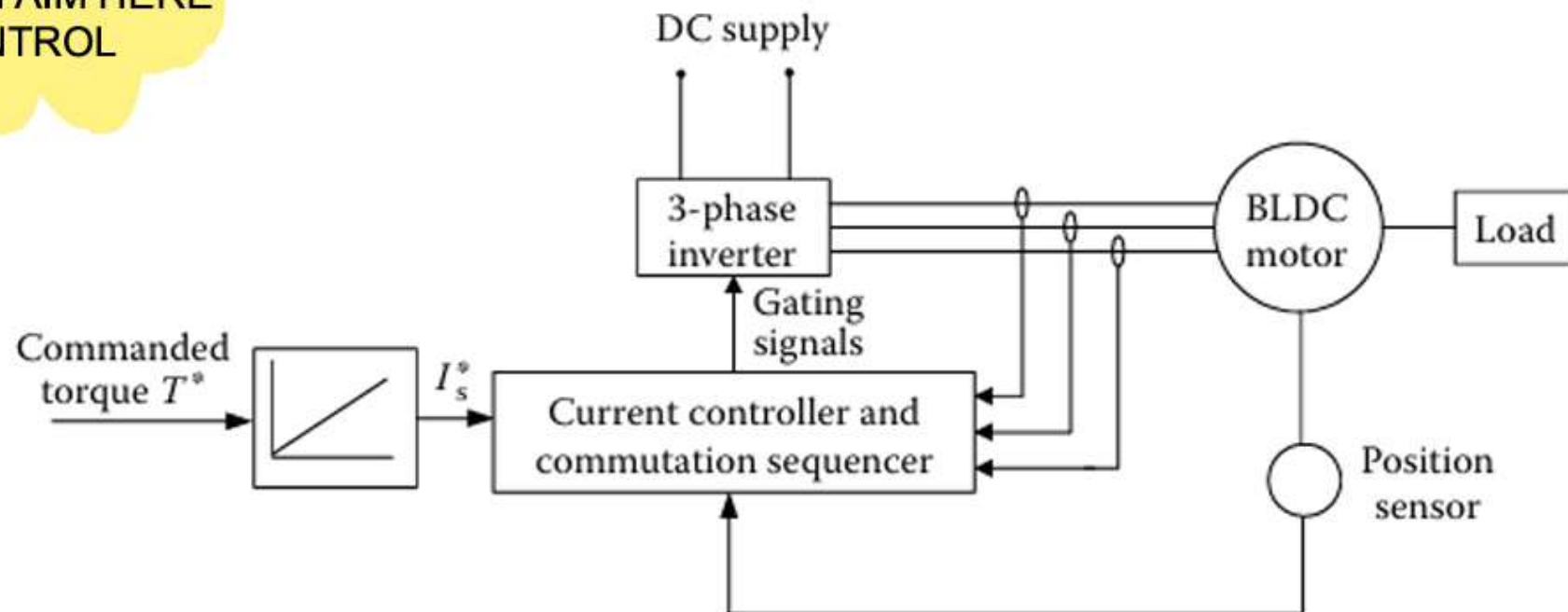
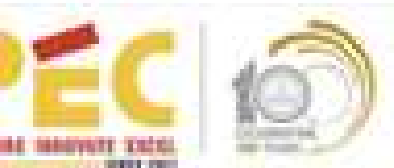


Figure 9: Block diagram of the torque control of the BLDC motor.

Figure 9 shows a block diagram of a torque control scheme for a BLDC motor drive. The desired current I^* is derived from the commanded torque T^* through a torque controller. The current controller and commutation sequencer receive the desired current I^* , position information from the position sensors, and perhaps the current feedback through current transducers, and then produce gating signals. These gating signals are sent to the three-phase inverter (power converter) to produce the desired phase current to the BLDC machine.

(see Figure 10). Many high-performance applications include current feedback for torque control. At the minimum, DC bus current feedback is required to protect the drive and machine from over-currents. The controller blocks, “speed controller” may be any type of classical controller such as a proportional-integral (PI) controller or a more advanced controller such as an artificial intelligence control. The “current controller and commutation sequencer” provides the properly sequenced gating signals to the “three-phase inverter” while comparing sensed currents to a reference to maintain a constant peak current control by hysteresis (current chopping) or with a voltage source (PWM)-type current control. Using position information, the commutation sequencer causes the inverter to “electronically commute,” acting as the mechanical commutator of a conventional DC machine. The commutation angle associated with a brushless motor is normally set so that the motor will commute around the peak of the torque angle curve. Considering a three-phase motor, connected in Delta or wye, commutation occurs at electrical angles, which are plus or minus 30° (electrical) from the peaks of the torque-angle curves. When the motor position moves beyond the peaks by an amount equal to



ELECTRIC VEHICLES (EEN5003)

DR. VENU SANGWAN



30° (electrical), then the commutation sensors cause the stator phase excitation to switch to move the motor suddenly to -30° relative to the peak of the next torque-angle curve

30° (electrical), then the commutation sensors cause the stator phase excitation to switch to move the motor suddenly to -30° relative to the peak of the next torque-angle curve

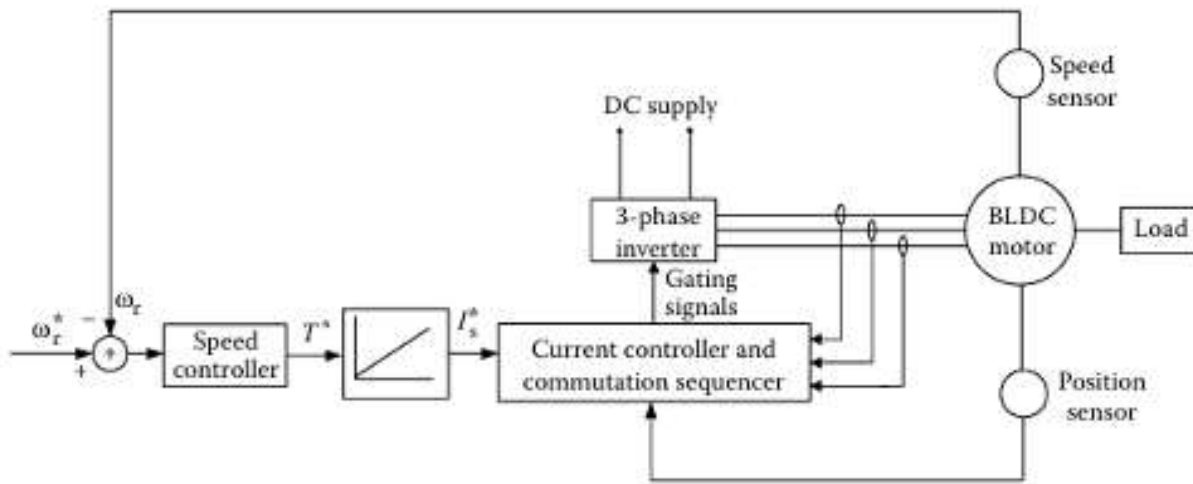


Figure 10: Block diagram of the speed control of the BLDC motor.

EXTEND SPEED TECHNOLOGY

As discussed above, PM BLDC machines inherently have a short constant power range due to their rather limited field weakening capability. This is a result of the presence of the PM field, which can only be weakened through production of a stator field component, which opposes the rotor magnetic field. The speed ratio, α , is usually less than 2.

Recently, the use of additional field windings to extend the speed range of PM BLDC motors has been developed. The key is to control the field current in such a way that the air gap field provided by PMs can be weakened during high-speed, constant-power operation. Due to the presence of both PMs and the field windings, these motors are the so-called PM hybrid motors. The PM hybrid motor can achieve a speed ratio of around 4. The optimal efficiency profiles of a PM hybrid motor drive are shown in Figure 11. However, the PM hybrid motors have the drawback of a relatively complex structure. The speed ratio is still not enough to meet the vehicle performance requirement, especially to an off-road vehicle. Thus, a multigear transmission is required.

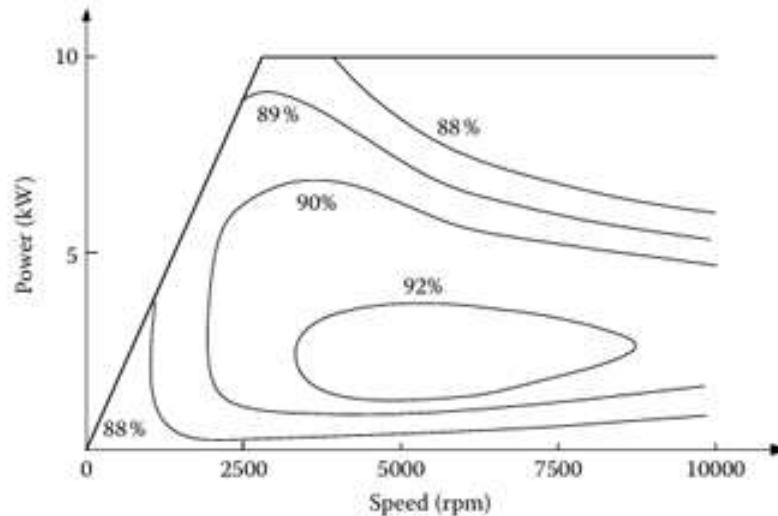


Figure 11: Optimal efficiency profiles of a PM hybrid motor drive

SENSORLESS TECHNIQUES

As mentioned above, the operation of the BLDC motor drives relies mostly on position sensors for obtaining the rotor position information so as to properly perform the turn-on or turn-off of each phase properly. The position sensor is usually either a three-element Hall-effect sensor or an optical encoder. These position sensors are high-cost, fragile elements. Thus, its presence not only enhances the cost of the motor drive but also seriously lowers its reliability and limits its application in some environments, such as the military one.

Position sensorless technology can effectively continue the operation of the system in case the position sensors lose their function. This is crucial in some applications, such as military vehicles. Several sensorless technologies have been developed. The majority of them are based on the voltage, current, and back EMF detection. These techniques can be primarily grouped into four categories:

1. Those using measured currents, voltages, fundamental machine equations, and algebraic manipulations.
2. Those using observers.
3. Those using back EMF methods.
4. Those with novel techniques not falling into the previous three categories.